AI-DRIVEN EARLY-WARNING SYSTEM FOR HUMAN-WILDLIFE COEXISTENCE

A PROJECT REPORT

Submitted by

MADHU ASMITHA P S – 921321243027

RAJA SUDHARSHANA GOVIND R – 921321243044

PARTHA SARATHY C A – 921321243040

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY,

(An Autonomous Institution Affiliated to Anna University, Chennai)

DINDIGUL-624622

MAY 2025

PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY,

(An Autonomous Institution Affiliated to Anna University, Chennai)

DINDIGUL-624622

BONAFIDE CERTIFICATE

Certified that this project report **"AI-DRIVEN EARLY-WARNING SYSTEM FOR HUMAN-WILDLIFE COEXISTENCE",** is the bonafide work of **MADHU ASMITHA P S (921321243027), RAJA SUDHARSHANA GOVIND R (921321243044), PARTHA SARATHY C A (921321243040),** who carried out the project under my supervision.

SIGNATURE

Dr. T. HEMALATHA, M.E., Ph.D.,

HEAD OF THE DEPARTMENT

Professor,

Department of AI & DS,

PSNA college of Engg & Tech

Dindigul- 624622

SIGNATURE

Mrs. G. JANANI @ PANDEESWARI, M.E

SUPERVISOR

Assistant Professor,

Department of AI & DS,

PSNA college of Engg & Tech

Dindigul-624622

Submitted for the Project Viva-Voce Examination held on ……

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

We thank the most graceful creator of the universe, our almighty **GOD** who ideally supported us throughout this project.

At this moment of having successfully completed our project, we wish to convey our sincere thanks to our **Chairperson Tmt. K. Dhanalakshmi Ammal** and our **Pro Chairman Rtn. MPHF R.S.K Raguraam, D.A.E M.Com,** for providing all the facilities to us

We would like to express our sincere thanks to our **Principal Dr. D.Vasudevan., M.E., Ph.D,** for letting us to do our project and offeringadequate duration in completing our project.

We would like to express our gratitude to our Head of Department of Artificial Intelligence and Data Science, **Dr.T.HEMALATHA, M.E., Ph.D,** for her constructive suggestions during our project.

With deep sense of gratitude, we extend our sincere thanks to our project guide **Mrs.G.JANANI @ PANDEESWARI, M.E.,** Assistant Professor, Department of Artificial Intelligence and Data Science, who is our light house in the vastocean of learning with her inspiring guidance and encouragement to complete the project. We would like to express our gratitude to our project coordinator by **Dr.P.RAMYA, M.E., Ph.D.,** Assistant Professor, Department of Artificial Intelligence and Data Science, for her valuable support throughout the project. Finally, we also offer our thanks to all teaching and non –teaching staff members of Artificial Intelligence and Data Science department and our friends for their kind help extended to us.

Madhu Asmitha P S

Raja Sudharshana Govind R

Partha Sarathy C A

ABSTRACT

Tribal communities in remote mountainous regions frequently face threats from wild animals, threatening their safety, food security, and economic stability. To counter this, our AI-powered early-warning system integrates advanced sensing technologies, intelligent deterrence mechanisms, sustainable energy solutions, and seamless communication networks. The system employs thermal infrared sensors to detect heat signatures, LiDAR sensors to precisely measure distance, speed, and size, and acoustic sensors to analyze vocalizations for species identification. AI-driven models, including Convolutional Neural Networks (CNN) and YOLO (You Only Look Once), process and classify animal sounds and images in real-time, reducing false alarms and ensuring accurate threat assessment. Upon detection, deterrent mechanisms such as ultrasonic sound emitters and high-intensity flashing lights activate strategically to repel animals without causing harm while optimizing energy use. The entire system is powered by solar energy with backup storage to ensure continuous operation, making it a sustainable and reliable solution even in remote areas. To enhance connectivity, a hybrid mesh network combining Wi-Fi and LoRa technology enables uninterrupted sensor communication across rugged terrains, facilitating swift data transmission. Real-time alerts are delivered via SMS notifications, and on-site alarms, allowing residents to take immediate precautions and connect with external agencies when needed. This Systematic approach not only enhances community safety and prevents wildlife-induced losses but also fosters a sustainable and ethical coexistence between humans and nature by leveraging cutting-edge AI, renewable energy, and resilient communication frameworks.

Keywords: LiDAR, YOLO, mesh network, CNN, LoRA

TABLE OF CONTENTS

CHAPTER NO.	TOPIC	PAGE NO.
	ABSTRACT	**iv**
	LIST OF FIGURES	**ix**
	LIST OF TABLES	**x**
	LIST OF ABBREVIATIONS	**xi**
1	**INTRODUCTION**	**1**
	1.1 BACKGROUND AND MOTIVATION	1
	1.2 CHALLENGES IN TRIBAL AND WILDLIFE SAFETY	2
	1.3 ROLE OF TECHNOLOGY IN THREAT DETECTION	2
	1.4 OBJECTIVES OF THE STUDY	3
2	**LITERATURE REVIEW**	**5**
3	**PROPOSED SYSTEM**	**7**
	3.1 SYSTEM ARCHITECTURE OVERVIEW	7
	3.2 KEY COMPONENTS OF THE SYSTEM	10

3.2.1 Sensing Unit 10

3.2.1.1 Thermal Imaging Cameras 10

3.2.1.2 LiDAR Sensors 11

3.2.1.3 Acoustic Sensors 11

3.2.2 Processing and Monitoring Unit 12

3.2.2.1 AI-Powered Wildlife Classification 12

3.2.2.2 Edge AI Computing 13

3.2.2.3 Threat Level Assessment 14

3.2.3 Communication Unit 14

3.2.3.1 Wi-Fi Communication 15

3.2.3.2 LoRa Wireless Network 15

3.2.4 Alert & Deterrent Layer 16

3.2.4.1 Auditory Deterrents 17

3.2.4.2 Visual Deterrents 17

3.2.4.3 Mobile Notifications 18

3.2.4.4 Automated Deterrent Activation 18

3.2.5 Energy Management 19

3.2.5.1 Solar Power Integration 19

3.2.5.2 Battery Backup System 19

3.2.5.3 Sustainable and Reliable Operation 20

4 IMPLEMENTATION AND RESULTS 21

4.1 SYSTEM SPECIFICATION 22

4.2 SOFTWARE AND HARDWARE COMPONENTS 23

4.2.1 Embedded Programming for Sensors 23

4.2.2 AI Model Development for Threat Classification 25

4.2.3 Thermal Image Classification 25

4.2.4 Audio-Based Classification 27

4.2.5 Communication Module Implementation 29

4.2.6 Alert Mechanism 32

4.3 DATASET DESCRIPTION 34

4.3.1 Thermal Image Dataset for Visual 35

Threat Detection

4.3.2 Audio Dataset for Acoustic Threat Recognition 37

4.3.3 Dataset Optimization for Edge Deployment 39

4.4 TESTING & VALIDATION 40

4.4.1 Unit Testing 41

4.4.2 Integration Testing 44

4.4.3 Field Deployment and Evaluation 46

4.4.4 Testcase 48

5 CONCLUSION AND FUTURE ENHANCEMENT 52

5.1 CONCLUSION 52

5.2 FUTURE ENHANCEMENTS AND SCALABILITY 53

REFERENCES 55

LIST OF FIGURES

FIGURE NO.	FIGURE NAME	PAGE NO.
3.1	System Architecture	7
4.1	Thermal image dataset	37
4.2	MFCC audio features dataset table	38
4.3	Thermal image detection	42
4.4	Audio detection	43
4.5	Incorrect image detection	49
4.6	Thermal image detection	50
4.7	Audio detection	51

LIST OF TABLES

TABLE NO.	TABLE NAME	PAGE NO.
4.1	System specification	22
4.2	Dataset Summary for Thermal and Audio Classification	35
4.3	YOLO Model Performance Metrics on Wildlife Detection Dataset	36

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
YOLO	You Only Look Once
CNN	Convolutional Neural Network
IoT	Internet of Things
LiDAR	Light Detection and Ranging
LoRa	Long Range
MFCCs	Mel-Frequency Cepstral Coefficients
MEMS	Micro-Electro-Mechanical Systems
ESP 32	Espressif Systems 32
LED	Light Emitting Diode
YAML	Yet Another Markup Language
CSV	Comma Separated Values
SMS	Short Message Service
GPIO	General Purpose Input/Output

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

Human-wildlife conflict is a growing concern, particularly for tribal communities living in remote forested regions. These communities often face severe threats from wild animals, such as elephants, leopards, wild boars, and sloth bears, which invade farmlands, destroy crops, and sometimes pose direct dangers to human lives. Unauthorized human activities, like poaching, illegal logging, and encroachment, put immense pressure on fragile ecosystems, disturbing the natural balance and often sparking more frequent and dangerous conflicts between people and wildlife.Tamil Nadu, with its extensive protected areas like Mudumalai Tiger Reserve and Sathyamangalam Wildlife Sanctuary, is one such region where frequent interactions between wildlife and indigenous tribes, such as the Irula, Toda, and Kurumba, create significant safety and livelihood challenges.

Traditional mitigation measures, such as manual patrolling and physical barriers, have proven ineffective in addressing the complexity of modern-day human-wildlife interactions. The unpredictability of animal movements and the limitations of conventional deterrent methods further aggravate the situation. Given these challenges, there is an urgent need for an AI-driven safety system that can offer real-time detection, predictive threat analysis, and instant communication alerts to protect vulnerable communities and ensure their harmonious coexistence with wildlife.

1.2 CHALLENGES IN TRIBAL AND WILDLIFE SAFETY

Despite efforts to manage human-wildlife conflicts, tribal communities in remote areas continue to face persistent challenges that make it difficult to ensure their safety and protect their livelihoods. The absence of real-time monitoring systems makes it nearly impossible for communities to be aware of approaching dangers until it is too late. Current methods such as manual patrolling and camera traps fail to provide instant alerts, limiting their ability to prevent attacks or crop destruction. Traditional deterrents like fire torches, loud noises, and electric fences have also proved ineffective over time, as many animal species adapt to them, rendering them less effective. Moreover, the high maintenance costs of physical barriers and their potential to harm both humans and animals make them unsustainable solutions.

Many current wildlife management strategies still depend on manual tracking and outdated data, making it challenging for authorities to take proactive steps. Addressing these challenges requires a technological intervention that can provide continuous monitoring, automated threat detection, and instant communication mechanisms to protect vulnerable communities.

1.3 ROLE OF TECHNOLOGY IN THREAT DETECTION

Advancements in Artificial Intelligence (AI), the Internet of Things (IoT), and sensor-based technologies have paved the way for innovative solutions in wildlife monitoring and human safety. AI-driven surveillance systems can significantly enhance threat detection capabilities, classify potential dangers, and provide timely alerts to communities and authorities. Deep learning models such as Convolutional Neural Networks (CNN) and You Only Look Once (YOLO) can accurately classify animal species based on thermal images captured by high-resolution thermal cameras. Acoustic sensors can further analyze sounds to

identify species-specific calls, helping to distinguish between non-threatening and dangerous wildlife.

Sensor-based monitoring technologies, such as thermal infrared sensors, enable the detection of animal heat signatures even in low-visibility conditions, ensuring 24/7 surveillance. LiDAR (Light Detection and Ranging) technology measures the distance, speed, and movement patterns of approaching animals, allowing for precise tracking of their behavior. IoT-enabled alert systems facilitate real-time data transmission between sensors and a centralized system, ensuring that communities receive timely notifications through mobile apps, text messages, and alarm systems. Additionally, automated deterrents such as ultrasonic sound emitters and high-intensity flashing lights can be activated based on real-time threat detection, optimizing energy efficiency while minimizing disruptions to non-target species. By integrating these technologies, it becomes possible to develop a robust and scalable system that enhances situational awareness, minimizes human casualties, and reduces economic losses due to wildlife invasions.

1.4 OBJECTIVES OF THE STUDY

The primary objective of this study is to design and implement an AI-driven early-warning system that enhances the safety and security of tribal communities while promoting sustainable human-wildlife coexistence. The research aims to design an automated system that uses AI to quickly detect and classify wildlife threats in real time, helping to reduce conflicts and enhance safety for both humans and animals recognition, acoustic analysis, and sensor data. Additionally, it seeks to design intelligent deterrents that activate only when necessary, reducing energy consumption and increasing their effectiveness in repelling animals without causing harm.

An IoT-based communication and alert system will be established using Wi-Fi and LoRa (Long Range) technology to ensure real-time alerts reach affected communities and authorities, even in remote locations with poor connectivity. To ensure uninterrupted operation in off-grid areas, the system will integrate renewable energy sources such as solar panels with battery backups. Furthermore, the proposed solution will be designed to be adaptable and scalable, enabling its implementation in different ecological regions and wildlife habitats . By addressing these objectives, this study aims to contribute to the development of a technologically advanced and socially impactful solution that safeguards both human lives and wildlife populations.

CHAPTER 2

LITERATURE REVIEW

S. P. Sahane, N. S. Salunke, V. S. Shirsath, S. S. Shukla, and A. R. Panhalkar (2019) [1] proposed an early warning system for detecting harmful animals using IoT. Their system uses sensor networks to monitor animal presence in real-time and trigger alerts to nearby human settlements. This approach relies on motion detection and environmental sensing to identify potential threats, allowing proactive measures to be taken before conflicts arise. The system emphasizes the importance of real-time data acquisition and timely intervention, reducing the risk of human-wildlife encounters.

R. Upadrashta et al. (2016) [2] developed a sensor array for intruder classification using PIR (Passive Infrared) sensors, integrated with animation and chirplet analysis. The system is designed for outdoor environments, focusing on differentiating between human and animal intruders. By capturing motion signatures and classifying them based on their thermal profiles, the system enhances the accuracy of wildlife threat detection and minimizes false alarms. The researchers highlight the critical role of sensor networks in improving the efficiency of early warning systems.

G. K. Verma and P. Gupta (2018) [3] presented a wild animal detection model using deep convolutional neural networks (CNN) at the 2nd International Conference on Computer Vision and Image Processing. Their approach leverages the power of deep learning to classify wildlife species based on camera trap images. By training their model on a large dataset of animal images, they achieved high accuracy in detecting various species, demonstrating the potential of AI in wildlife conservation. The study underscores the effectiveness of computer vision techniques in automating wildlife monitoring.

A. P. N. Achyutha et al. (2023) [4] conducted an investigation into forest fire detection, wildlife preservation, and tribal preparedness using deep learning and IoT. Their work explores the use of sensor networks and real-time data analytics to improve the safety of forested regions. The study combines multiple data sources, including temperature, humidity, and motion sensors, to provide comprehensive early warning systems that address both wildlife threats and environmental hazards.

M. Knyva et al. (2023) [5] proposed an IoT sensor network for wild animal detection near roads, aimed at reducing roadkill and enhancing driver safety. Their system uses edge devices to monitor animal movements near highways, transmitting real-time data to traffic management systems. This approach not only reduces the risk of accidents but also contributes to wildlife conservation by minimizing habitat fragmentation and reducing animal mortality.

H. Patil and N. Ansari (2021) [6] introduced an automated wild animal intrusion detection and repellent system using artificial intelligence of things (AIoT) at the 4th International Conference on Advanced Science and Technology. Their model integrates AI with IoT devices to detect and deter animals approaching human settlements. By combining real-time detection with automated deterrents, this system provides an innovative solution to human-wildlife conflicts, ensuring both human safety and wildlife conservation.

C. Panagiotopoulos (2024) [7] explored machine learning for animal detection and recognition in his M.S. thesis at the Hellenic Open University. His work focused on the application of AI algorithms to accurately identify wildlife species in real-time, providing valuable insights into habitat use and movement patterns. This approach supports conservation efforts by enhancing the ability to monitor and protect endangered species.

CHAPTER 3

PROPOSED SYSTEM

3.1 SYSTEM ARCHITECTURE OVERVIEW

The AI-Driven Early-Warning System for Human-Wildlife Coexistence is an advanced, multi-layered architecture that integrates real-time sensing, intelligent processing, and efficient communication to mitigate human-wildlife conflicts. This system is built to function in remote, forested, and low-infrastructure areas, where traditional monitoring methods fail due to limited connectivity and power supply. The proposed framework consists of five interconnected layers, ensuring seamless wildlife detection, rapid classification, proactive deterrence, and efficient energy management. The Detection Layer is responsible for continuous environmental surveillance using thermal imaging cameras, LiDAR sensors, and acoustic monitoring units.

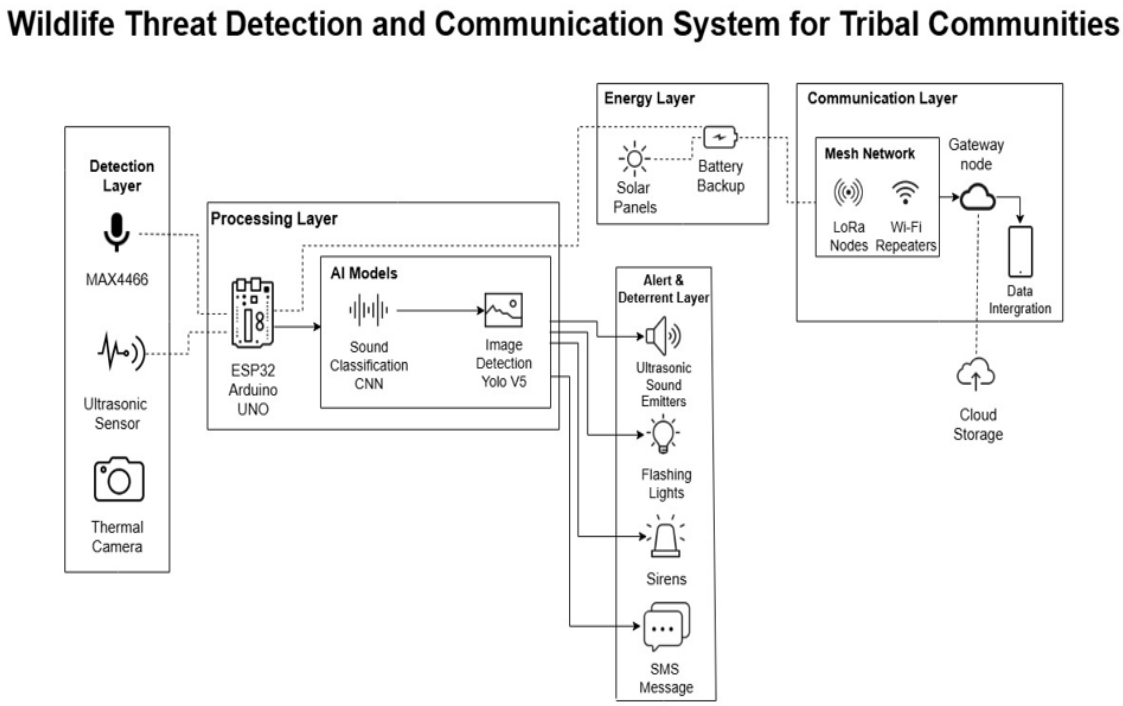


Fig.3.1 System Architecture

Thermal cameras detect heat signatures of animals, even in low-light conditions or dense vegetation, making them an essential tool for nighttime monitoring. LiDAR sensors generate detailed spatial maps by measuring the distance and movement of objects, ensuring precise tracking of wildlife movements, even in uneven terrains. Acoustic sensors analyze animal vocalizations to identify species presence and behavioral patterns. These sensors work together to minimize false alarms and improve detection accuracy by providing multi-modal data inputs. Once the raw data is gathered, it is processed in the Processing Layer, where AI-powered algorithms classify species and assess the level of threat posed by the detected wildlife. Convolutional Neural Networks (CNNs) extract patterns from Mel-Frequency Cepstral Coefficients (MFCC) features of animal vocalizations, enabling precise species identification. The YOLO (You Only Look Once) model is employed for real-time image-based wildlife detection, allowing rapid identification of animals detected through thermal and LiDAR cameras. This edge computing approach, using Raspberry Pi ensures that critical decisions are made locally, reducing reliance on cloud-based computing and preventing latency in emergency situations. The Threat Classification Decision Nodes evaluate species type, proximity to human settlements, and behavioral patterns, determining whether an immediate deterrent response is required. To facilitate seamless communication in challenging terrains, the Communication Layer implements a hybrid network approach utilizing LoRa (Long Range) wireless technology, Wi-Fi repeaters. LoRa networks enable long-distance data transmission with minimal power consumption, ensuring reliable connectivity even in deep forest regions where cellular networks may be unavailable. Wi-Fi modules provide high-speed data transfer for local monitoring stations, facilitating real-time streaming of camera feeds and acoustic signals. A central gateway node aggregates and transmits data to a cloud platform, where it can be accessed by wildlife authorities, conservationists, and tribal communities. Additionally, SMS alerts ensure that

villagers receive immediate warnings if dangerous wildlife is detected nearby. The Alert & Deterrent Layer is designed to prevent wildlife from encroaching into human settlements while ensuring their safety. Upon detecting a threat, the system activates deterrent mechanisms tailored to the species and risk level. Auditory deterrents, such as ultrasonic sound waves and pre-recorded distress calls, scare animals away without causing harm. Visual deterrents, including flashing LED lights and reflective materials, create a disorienting effect to prevent wildlife from advancing further. Additionally, real-time alerts are sent to local authorities, forest officials, and affected communities via SMS and mobile notifications, allowing them to take necessary precautions. The AI system ensures optimal deterrent activation by analyzing wildlife behavior, time of day, and environmental conditions, minimizing unnecessary disturbances to animals and humans. Given that the system is deployed in remote locations, the Energy Layer incorporates renewable energy solutions for uninterrupted operation. Solar panels provide a sustainable, off-grid power source, while battery backup systems ensure that operations continue during nighttime and cloudy weather conditions. AI-driven energy management algorithms optimize power consumption, activating high-energy deterrent mechanisms only when necessary to conserve power and extend battery life. By leveraging self-sustaining energy solutions, the system remains cost-effective, environmentally friendly, and operationally independent of conventional power grids. This AI-driven system offers a scalable, real-time, and proactive solution to human-wildlife conflicts. By combining advanced sensing technologies, deep learning models, and efficient communication networks, it significantly enhances wildlife detection, risk assessment, and threat mitigation. The multi-layered architecture ensures that local communities receive timely alerts, enabling them to take preventive actions, protect livelihoods, and promote peaceful human-wildlife coexistence. Ultimately, this system serves as a technological bridge between conservation efforts and human safety, fostering sustainable wildlife management while preserving ecological balance.

3.2 KEY COMPONENTS OF THE SYSTEM

The system comprises five core modules, each performing a critical role in enhancing wildlife monitoring and improving human safety.

3.2.1 Sensing Unit

The Sensing Unit is the foundation of the system, responsible for gathering real-time environmental data to detect, classify, and track wildlife activity. By integrating multiple sensor technologies, this unit ensures accurate detection even in challenging terrains and lighting conditions. The combined use of thermal imaging cameras, LiDAR sensors, and acoustic monitoring devices allows for a multi-modal approach to wildlife surveillance, significantly enhancing the reliability of early warning alerts.

3.2.1.1 Thermal Imaging Cameras

Thermal imaging cameras play a crucial role in wildlife detection, especially in low-light conditions or areas with dense vegetation where standard optical cameras may fail. These cameras rely on infrared sensors to detect the heat signatures emitted by animals, enabling surveillance even in complete darkness. By leveraging AI-driven image processing models, the system can differentiate between various animal species and non-threatening objects such as human activities, vehicles, or environmental obstacles. The integration of YOLO-based real-time object detection ensures that the system not only identifies wildlife but also assesses their proximity and movement patterns. Additionally, heat anomaly detection algorithms can recognize unusual thermal patterns, such as distressed animals or large herds moving together, further improving the system’s effectiveness in providing early warnings.

3.2.1.2 LiDAR Sensors

LiDAR (Light Detection and Ranging) sensors provide an additional layer of precision by generating high-resolution 3D spatial maps of the monitored environment. Unlike conventional cameras, which may struggle in areas with thick vegetation or uneven landscapes, LiDAR sensors use laser pulses to measure distances, detect object sizes, and track movement speeds. This allows for accurate mapping of wildlife movement patterns, particularly in terrains where visual surveillance is limited. AI-powered data processing using CNN and YOLO analyzes the spatial information, distinguishing between stationary objects, moving wildlife, and external disturbances such as falling branches or shifting shadows. By integrating LiDAR with thermal imaging, the system ensures a comprehensive, high-accuracy detection network, enabling conservationists and local authorities to predict animal movement and mitigate potential human-wildlife encounters effectively.

3.2.1.3 Acoustic Sensors

Acoustic monitoring adds another crucial dimension to the detection system by analysing wildlife sounds. This method is especially valuable in dense forests, where visibility is often compromised, making it difficult for traditional imaging technologies to operate efficiently. The acoustic sensors continuously capture environmental sounds and use AI-driven machine learning models, specifically CNN-based audio classifiers trained on Mel-Frequency Cepstral Coefficients (MFCC) features, to identify different animal species based on their unique sound signatures.

By converting raw audio signals into frequency-based spectrograms and analysing variations in pitch, duration, and frequency patterns, the AI system can distinguish between species with high accuracy. For instance, the model can differentiate between the growl of a tiger, the trumpet of an elephant, and the

distress call of a deer, each of which indicates different levels of potential threat. Furthermore, the system can identify abnormal sound patterns, such as an increase in predator activity or distress calls from prey animals, which may signal an imminent danger to humans or livestock nearby. By combining thermal imaging cameras, LiDAR sensors, and acoustic monitoring, the system ensures highly accurate wildlife tracking while minimizing false alarms. The multi-sensor approach enhances early-warning capabilities, allowing communities, conservationists, and authorities to respond proactively to wildlife presence while ensuring human safety and ecological balance.

3.2.2 Processing and Monitoring Unit

The Processing and Monitoring Unit serves as the system's intelligence hub, handling real-time data analysis and automating response mechanisms. By leveraging advanced AI-driven models and edge computing, this unit ensures rapid decision-making while reducing dependency on cloud infrastructure. The seamless integration of AI-powered wildlife classification, edge AI computing, and threat level assessment allows for a highly responsive and autonomous early-warning system.

3.2.2.1 AI-Powered Wildlife Classification

To accurately identify and classify wildlife species, the system employs deep learning models, specifically Convolutional Neural Networks (CNNs) for analysing acoustic sensor data and YOLO (You Only Look Once) for real-time image-based species recognition. The CNN-based audio processing model extracts Mel-Frequency Cepstral Coefficients (MFCC) features, enabling precise identification of species based on their vocalizations. By detecting unique frequency patterns, the system can differentiate between different animal calls, such as predatory growls, warning signals, and distress calls, providing crucial insights into wildlife presence and activity.

Meanwhile, the YOLO deep-learning model processes real-time visual data from thermal cameras and LiDAR sensors to recognize and classify animals. Unlike conventional object detection methods, YOLO enables instantaneous species recognition by processing images in a single pass, ensuring that wildlife can be identified in milliseconds. This rapid classification allows for real-time monitoring and enhances the system's ability to differentiate between threatening wildlife (such as predators) and non-threatening species. By integrating these AI-based models, the system achieves high accuracy in species classification, reducing false alarms and enabling more precise threat assessment.

3.2.2.2 Edge AI Computing

To improve processing efficiency and ensure real-time responsiveness, the system incorporates edge AI computing, where sensor data is processed locally using low-power AI devices such as Raspberry Pi. Instead of relying on continuous cloud connectivity, edge AI devices analyse incoming data on-site, drastically reducing latency and enabling instant decision-making. This is particularly crucial in remote areas with limited or no internet access, where real-time processing can determine the effectiveness of wildlife deterrent mechanisms. By performing computations closer to the source, edge AI computing also enhances system reliability and energy efficiency. Large volumes of sensor data—thermal images, LiDAR maps, and acoustic recordings—are processed locally, with only critical alerts and summarized data sent to cloud storage for long-term analysis. This decentralized approach not only reduces bandwidth usage but also minimizes the risk of communication failures, ensuring continuous monitoring and response activation.

3.2.2.3 Threat Level Assessment

To determine the severity of wildlife encounters, the system employs an AI-driven risk evaluation model that assigns a threat score based on multiple factors, including animal behaviour, species classification, movement patterns, and proximity to human settlements. The threat score is dynamically adjusted based on real-time observations. For example, a lone grazing animal near a forest edge may receive a low threat score, whereas a predatory animal approaching a village would trigger a high-alert status.

When the threat level crosses a predefined threshold, the system automatically activates deterrent mechanisms such as ultrasonic sound emitters, flashing lights, and sirens to repel the wildlife before conflicts occur. Simultaneously, automated SMS are sent to local authorities, conservationists, and nearby communities, allowing them to take precautionary measures. The system can also be configured to send alerts to forest rangers, and tribal people ensuring that potential threats are mitigated before escalation. By integrating AI-based wildlife classification, edge AI computing, and intelligent threat assessment models, the Processing and Monitoring Unit guarantees fast, efficient, and real-time decision-making, significantly improving wildlife monitoring and human-wildlife conflict prevention.

3.2.3 Communication Unit

The Communication Unit plays a vital role in ensuring seamless data transmission between various system components while delivering timely alerts to authorities and local communities. Given that wildlife monitoring often occurs in remote and inaccessible regions, the system implements a hybrid communication strategy that integrates Wi-Fi, LoRa (Long-Range) wireless networks, cloud integration, and mobile-based alert mechanisms. This multi-

layered approach ensures real-time threat detection, continuous data accessibility, and instant notifications, reducing human-wildlife conflicts effectively.

3.2.3.1 Wi-Fi Communication

The system leverages Wi-Fi technology for high-speed data transmission between sensors, AI processing units, and monitoring stations. With the ability to support real-time video streaming, audio processing, and LiDAR-based terrain mapping, Wi-Fi ensures that all collected sensor data is instantly available for analysis. This connectivity allows researchers, security personnel, and AI processing systems to monitor wildlife activity in real-time, enabling immediate response actions.

Additionally, Wi-Fi networks facilitate continuous monitoring through live video feeds from thermal cameras, allowing conservationists to observe animal movements remotely. However, due to its limited range, Wi-Fi is primarily used for short-range high-bandwidth communication within specific areas such as wildlife reserves, buffer zones, and community monitoring centers.

3.2.3.2 LoRa (Long-Range) Wireless Network

To overcome the distance limitations of Wi-Fi and ensure reliable connectivity in vast landscapes with minimal infrastructure, the system employs LoRa (Long-Range) wireless communication. LoRa technology is designed for low-power, long-distance transmission, making it ideal for wildlife monitoring in forests, grasslands, and mountainous regions. The system deploys LoRa-enabled sensor nodes, which continuously transmit wildlife alerts over a range of several kilometers without requiring cellular or internet connectivity.

LoRa ensures effective communication between remote sensing units and centralized monitoring stations, even in areas where mobile networks or internet access is unreliable or unavailable. By enabling long-distance data transfer with

minimal power consumption, LoRa enhances the system's sustainability, ensuring that wildlife detection and classification alerts can reach monitoring centers and local communities without interruption.

Given that not all communities have smartphones or internet access, the system also utilizes SMS-based notifications to broadcast emergency alerts. In the event of a high-threat detection—such as a predator approaching human settlements—the system automatically sends text messages to local authorities, and security personnel, enabling them to respond swiftly and safeguard both human and wildlife populations. By integrating Wi-Fi, LoRa, cloud technology, and mobile-based alert systems, the Communication Unit ensures continuous data flow, immediate response activation, and enhanced public safety. This hybrid approach minimizes communication delays and strengthens proactive measures, making it a critical component in achieving a sustainable and efficient human-wildlife coexistence system.

3.2.4 Alert & Deterrent Layer

The Alert & Deterrent Layer is a crucial component of the system that activates protective measures when a wildlife threat is detected, ensuring human safety while preventing harm to animals. This layer is designed to minimize human-wildlife conflicts by implementing a combination of sensory warnings, automated deterrents, and real-time communication. When the AI-powered monitoring system determines a potential threat, it intelligently deploys the most effective deterrent based on the species, threat level, and proximity to human settlements. These deterrents use a non-lethal approach to prevent wildlife from encroaching into residential areas while maintaining ecological balance.

3.2.4.1 Auditory Deterrents

Auditory deterrents play a significant role in keeping wildlife away from human settlements by leveraging sound-based scare tactics. The system uses pre-recorded distress calls of predators or high-frequency sirens that are known to be effective in repelling specific species. For example, elephants are highly sensitive to the sound of bees, and pre-recorded bee swarm noises can effectively prevent them from invading into residential. Similarly, ultrasonic sound waves—beyond human hearing but audible to certain animals—can serve as an effective deterrent for predatory species. These auditory deterrents are automatically triggered when a high-risk animal is detected approaching a village . The sound intensity and frequency can be adjusted dynamically based on the type of wildlife, ensuring an effective yet humane approach to deterrence.

3.2.4.2 Visual Deterrents

Visual deterrents enhance the system's ability to prevent wildlife encroachment by using bright, unpredictable light patterns to create a sense of confusion and disorientation. The system employs flashing LED lights that trigger an avoidance response in animals. Many nocturnal species, such as leopards and wild boars, are particularly sensitive to sudden bursts of light, making visual deterrents a powerful tool, especially during nighttime intrusions. These deterrents are strategically positioned around village boundaries, and forest peripheries to create an invisible barrier that discourages animals from proceeding further. The intensity, duration, and frequency of the light flashes are optimized based on the species detected to ensure effectiveness while minimizing excessive energy consumption.

3.2.4.3 Mobile Notifications

In addition to deploying active deterrents, the system ensures that local communities, wildlife authorities, and monitoring personnel receive instant alerts about potential wildlife threats. Mobile notifications are sent via SMS to individuals residing in affected areas, allowing them to take preventive measures. These alerts provide real-time updates on the type of wildlife detected, its exact location, and its estimated time of arrival near human settlements. The mobile application features an intuitive dashboard where users can access past wildlife activity records, enabling better preparedness for future encounters. The notification system is designed to prioritize high-risk threats, ensuring that relevant stakeholders receive critical alerts without unnecessary disruptions.

3.2.4.4 Automated Deterrent Activation

To optimize energy usage and ensure an effective response, the system employs AI-powered risk assessment models that determine the best deterrent strategy based on real-time data. Instead of activating all deterrents at once, the system intelligently selects the most appropriate method based on the type of animal, its movement speed, and its past behaviour patterns. For instance, if an elephant herd is detected moving towards residential area, the system may first activate auditory deterrents, followed by flashing lights if the animals continue approaching. This multi-layered, adaptive deterrent mechanism prevents habituation, where animals become desensitized to a single deterrent over time. Additionally, the system logs every deterrent activation event, allowing researchers and conservationists to analyse long-term wildlife movement patterns and improve the effectiveness of the intervention strategies.

By integrating auditory and visual deterrents with automated threat assessment and mobile notifications, this multi-layered alert mechanism significantly reduces human-wildlife conflicts while maintaining ecological

harmony. The system ensures that wildlife is gently redirected without harm, fostering a balanced coexistence between human communities and the natural environment.

3.2.5 Energy Management

In wildlife-dense buffer zones and forest borders, ensuring a stable and uninterrupted power supply is critical for the functionality of any remote threat detection system. The absence of conventional power infrastructure in these regions necessitates a self-reliant, low-maintenance, and sustainable energy model to support the operation of sensors, cameras, microcontrollers, and alert mechanisms deployed in the system.

3.2.5.1 Solar Power Integration

To meet this requirement, the system integrates photovoltaic solar panels that serve as the primary source of energy. These panels harness solar radiation and convert it into electrical power, which is used to operate the ESP32 microcontroller, thermal and acoustic sensors, LoRa/WiFi communication modules, and alerting devices such as sirens, LEDs, and ultrasonic emitters. Solar energy provides a renewable, eco-friendly, and cost-effective solution, eliminating dependency on grid electricity and enabling deployment in remote and forested areas.

3.2.5.2 Battery Backup System

To ensure round-the-clock operation, especially during nighttime or cloudy days, the system is equipped with a battery backup module. This module consists of high-capacity lithium-ion or deep-cycle batteries that store excess power generated by the solar panels during daylight hours. The stored energy ensures continuous system operation even in the absence of direct sunlight. This backup

mechanism is vital to maintain the integrity and responsiveness of the system under all weather and light conditions.

3.2.5.3 Sustainable and Reliable Operation

The combination of solar panels and reliable battery storage ensures the system operates continuously and autonomously with minimal human intervention. This energy setup not only reduces maintenance overhead but also supports sustainable deployment in ecologically sensitive and inaccessible locations. By using renewable energy sources, the system aligns with environmental preservation goals while safeguarding tribal communities and forest ecosystems against potential wildlife threats.

CHAPTER 4

IMPLEMENTATION AND RESULTS

4.1 SYSTEM SPECIFICATION

The successful implementation of the proposed Tribal Safety and Wildlife Threat Detection System hinges on the precise definition and integration of both hardware and software components. This section lays the groundwork for understanding the technical architecture that enables real-time threat detection in forested and remote tribal areas. Given the unpredictable nature of wildlife movement and the lack of conventional infrastructure in such regions, the system must be designed to function autonomously, reliably, and with minimal human intervention. The specifications detailed herein ensure that the system can perform a wide range of tasks—from sensing and analyzing environmental inputs to transmitting alerts and managing power resources efficiently.

Each component has been selected with a focus on low power consumption, rugged durability, scalability, and interoperability with other modules. The hardware setup includes sensory devices such as thermal imaging cameras and acoustic microphones, supported by power systems like solar panels and battery backups, and communication interfaces like LoRa and Wi-Fi modules. On the software side, deep learning models, audio feature extraction algorithms, and embedded firmware coordinate to facilitate detection, classification, and alert generation. The software ecosystem is tailored to process both image and audio data formats, while ensuring real-time decision-making through optimized workflows. Collectively, these specifications serve as the technological backbone of the system, enabling it to operate efficiently in diverse environmental conditions and fulfill its objective of protecting tribal populations and preserving wildlife ecosystems.

Category	Component	Description
	Thermal Camera	Detects heat signatures for nocturnal and low-visibility conditions.
	AI Camera	Provides real-time image classification using deep learning.
Hardware	LiDAR Sensor	Measures distance and speed of moving objects for precise tracking.
	ESP32 Module	Provides Wi-Fi and Bluetooth connectivity for real-time data transmission and remote monitoring.
	Microphone	Captures animal vocalizations for sound-based detection.
	Power Supply	Provides necessary power for continuous operation.
Software	Python	Core programming language for AI and data processing.

	TensorFlow	Machine learning framework for model training and inference.
	Flask	Lightweight web framework for data APIs.
IoT Programming	Arduino IDE	Used for sensor programming and microcontroller integration.

Table 4.1 System Specification

4.2 SOFTWARE AND HARDWARE COMPONENTS

The successful implementation of the Tribal Safety and Wildlife Threat Detection System is highly dependent on the effective integration of both software and hardware elements. This section describes the tools, frameworks, programming methodologies, and communication protocols employed in the system. The platform combines sensor-based embedded systems with machine learning-based classification models, backed by communication technologies and real-time alerting infrastructure. Each component has been carefully selected and programmed to ensure high reliability. The system achieves end-to-end functionality—from sensing and detection to decision-making and response—through the process of integration of these components.

4.2.1 Embedded Programming for Sensors

Embedded programming plays a pivotal role in the seamless integration of sensors with the microcontroller, forming the foundational layer of data acquisition in the Tribal Safety and Wildlife Threat Detection System. The ESP32 microcontroller is chosen as the core processing unit due to its multi-protocol

capabilities, low power consumption, and compatibility with various sensor interfaces. Programming is carried out using Embedded C within environments such as the Arduino IDE, which provide direct access to the microcontroller's hardware resources, including GPIOs, ADCs, SPI, I2C, and UART communication channels.

For the thermal imaging camera (e.g., FLIR Lepton or Seek Thermal), interfacing is implemented through the SPI protocol, where each thermal frame is composed of a matrix of temperature readings corresponding to infrared intensities. These values are read in real-time and processed into a heat map structure. The code includes frame buffering, signal calibration, and temperature thresholding to identify the presence of warm-blooded creatures such as wild animals or human intruders.

Simultaneously, acoustic sensors, which may include condenser microphones or MEMS microphone arrays, are interfaced using the ESP32's ADC (Analog-to-Digital Converter) or I2S (Inter-IC Sound) peripherals. Real-time audio signal sampling is programmed at rates such as 16 kHz or 44.1 kHz, allowing for high-resolution acoustic pattern capture. The embedded firmware employs interrupt service routines (ISRs) to manage continuous audio sampling without data loss. Signal conditioning techniques such as filtering and normalization are implemented at the edge level to prepare the data for machine learning classification while reducing bandwidth requirements.

To conserve power, especially in off-grid forest environments, embedded routines are written to implement deep sleep and wake-up triggers based on motion detection, sudden sound bursts, or temperature anomalies. Additionally, all sensor interfacing code includes error-handling mechanisms, timing synchronization, and logging functionalities to ensure robust performance under variable environmental conditions. This low-level programming forms the

backbone of a responsive, intelligent, and energy-efficient wildlife monitoring system.

4.2.2 AI Model Development for Threat Classification

The intelligent capability of the proposed Tribal Safety and Wildlife Threat Detection System is fundamentally driven by its ability to accurately recognize and classify potential threats in dense forest regions and tribal settlements. The accuracy is vital for ensuring timely intervention and safeguarding human and animal life. To accomplish this, the system leverages advanced machine learning models that are meticulously developed and trained using two primary data modalities: thermal images and environmental audio signals. These sources provide complementary information - visual cues from heat signatures and auditory cues from wildlife sounds or human activity—enhancing the system's detection robustness in complex natural environments. By analyzing both types of data, the models can perform real-time classification of objects and sound patterns, allowing the system to issue alerts promptly and accurately. This section presents a detailed breakdown of how the thermal image classification model and the audio classification model are designed, trained, evaluated, and deployed for operational use in the field.

4.2.3 Thermal Image Classification

Thermal imaging is a cornerstone of the proposed system's visual threat detection mechanism, particularly effective in low-light, nighttime, or foggy conditions where conventional RGB cameras may fail. Unlike optical imagery, thermal cameras detect infrared radiation emitted by objects, converting it into a heat map representation that reveals the temperature distribution of living beings and the surrounding environment. This unique capability allows the system to detect and classify wildlife and human intruders based on their heat signatures, regardless of ambient light or weather conditions.

The dataset used for training the thermal image classification model is curated with infrared images captured at a standardized resolution of 640×640 pixels in JPG format. These images contain the thermal profiles of various entities commonly encountered in forested or tribal regions, including cheetahs, elephants, bears, and humans. Each image is paired with an annotation file following the YOLO (You Only Look Once) format. These text-based annotations include the class label and the bounding box coordinates (centre x, center y, width, height in normalized values) that specify the object's location in the image. A corresponding YAML configuration file is used to define key dataset parameters, such as the list of class names, number of object categories, and directory paths to training and testing datasets.

For real-time object detection and classification, the YOLOv5 architecture is employed due to its balance of speed and accuracy. YOLOv5 is known for processing images in a single forward pass, making it highly efficient for time-sensitive detection tasks in embedded systems. The architecture consists of multiple convolutional layers that extract hierarchical features from thermal input, followed by detection layers that output bounding box coordinates and class probability scores for each detected object. This one-stage detection process significantly reduces inference time and computational overhead, making it suitable for deployment on edge devices.

Model training is conducted using rigorous hyperparameter optimization to ensure high detection accuracy and generalizability. Parameters such as learning rate, number of epochs, batch size, and optimizer type are fine-tuned based on experimental results. To improve robustness and reduce overfitting, various data augmentation techniques are applied to the training dataset. These include horizontal flipping to simulate directional movement, brightness scaling to mimic environmental variations, and random cropping to introduce positional diversity. The model iteratively refines its weights over several epochs, gradually

learning to distinguish between the thermal patterns of different species and human intruders.

Once the model reaches optimal performance, it is exported in a lightweight format, allowing it to be integrated with edge computing platforms. Depending on the deployment configuration, the model can be interfaced via the ESP32 microcontroller with support from an external processing unit. This modularity allows the system to adapt to different energy and hardware constraints while maintaining real-time processing capabilities.

In essence, the thermal classification module offers a reliable and energy-efficient solution for automated wildlife and intruder detection, significantly enhancing the safety of remote tribal regions. Its ability to operate effectively under low-visibility conditions makes it an indispensable part of the overall threat detection architecture.

4.2.4 Audio-Based Classification

To complement the thermal imaging module and enhance the multi-modal threat detection capability of the proposed system, an acoustic sensing and classification layer is incorporated. This layer plays a critical role in detecting audio-based cues that may signal the presence of wildlife or other potential threats. Unlike visual detection—which may be limited by obstacles such as terrain, vegetation, or poor lighting—sound can propagate over long distances, providing a valuable early warning mechanism.

High-sensitivity microphones are strategically placed across the monitoring zones to continuously capture environmental audio signals such as animal calls, growls, roars, or footsteps. These sounds are recorded and stored in .wav format, preserving the full frequency spectrum needed for precise analysis.

Once the audio data is collected, Mel-Frequency Cepstral Coefficients (MFCCs) are extracted from each sample. MFCCs are widely used in audio

classification tasks because they capture key characteristics of sound, closely mimicking the human ear's perception of audio. For each sound clip, a fixed-length MFCC feature vector is computed—typically spanning 40 coefficients across a defined window of time—to create a compact and informative representation of the audio signal.

The resulting structured dataset is saved in a CSV format, where each row corresponds to a single .wav file. Each row contains the flattened MFCC feature vector and a corresponding class label such as "Lion", "Elephant", "Bear", "Buffalo", "Cheetah", "Dog", or "Goat". This labelled dataset forms the basis for training various machine learning models.

To classify the audio data, several algorithms can be employed:

- Convolutional Neural Networks (CNNs) are useful when MFCCs are reshaped into spectrogram-like matrices, allowing the model to capture spatial and temporal patterns in the sound.
- Support Vector Machines (SVMs) are effective for smaller datasets with well-defined class boundaries.
- Multi-Layer Perceptron's (MLPs) offer a straightforward and efficient architecture for learning from fixed-size feature vectors like flattened MFCCs.

These models learn to associate distinct MFCC patterns with specific animal species, enabling the system to perform real-time acoustic classification for threat detection and wildlife monitoring in diverse environments.

To ensure that the model generalizes well to unseen data, the dataset is split into training and testing subsets, typically following an 80:20 ratio. During training, models undergo several epochs of iterative optimization, often using techniques like early stopping or dropout to avoid overfitting. Additionally, further audio descriptors such as Chroma vectors (which capture harmonic content) and Spectral Contrast (which reflects the variance between peaks and

valleys in the spectrum) may be explored to improve classification performance, especially in noisy outdoor environments like forests where background interference is common.

Model evaluation is conducted using standard classification metrics such as accuracy, precision, recall, and F1-score, providing insights into how well the model distinguishes between threat categories. A confusion matrix is also generated to visualize classification performance across all classes, highlighting any common misclassifications that may need attention. These evaluations guide subsequent refinements in data preprocessing, feature selection, and model architecture.

In deployment, the selected audio classification model is converted to a lightweight format and integrated with the ESP32-based edge system. It may run locally on an attached low-power processing unit or on the ESP32 with optimized libraries, enabling real-time acoustic threat detection and prompt alert generation. This audio classification capability significantly boosts the reliability of the system, especially in scenarios where visual confirmation may not be possible, thus ensuring comprehensive monitoring of forest zones.

4.2.5 Communication Module Implementation

Reliable and continuous communication is a cornerstone of the proposed Tribal Safety and Wildlife Threat Detection System, especially considering the environmental and infrastructural constraints inherent to forested and tribal regions. The primary objective of the communication subsystem is to ensure that data generated by the system—such as real-time threat detections, sensor statuses, and alerts—can be transmitted seamlessly and promptly to centralized monitoring units or directly to mobile applications accessed by forest rangers and tribal users. To meet this requirement, the system adopts a hybrid communication architecture that combines LoRa (Long Range Radio) and Wi-Fi technologies. This dual-

layered communication strategy not only enhances the robustness and flexibility of data transmission but also ensures operational continuity even under variable network availability and terrain-induced connectivity limitations.

The LoRa (Long Range) module is central to the system's ability to communicate across vast distances without the need for cellular infrastructure. LoRa technology operates on low-power, license-free sub-GHz radio frequency bands (such as 433 MHz or 868 MHz), which are well-suited for rural and forest deployments due to their ability to penetrate foliage and rugged terrain. Within this system, the LoRa module is interfaced with the ESP32 microcontroller, using either SPI (Serial Peripheral Interface) or UART (Universal Asynchronous Receiver-Transmitter) protocols for data exchange. Configuration parameters such as spreading factor (SF), bandwidth, coding rate, and transmit power are carefully calibrated to balance transmission distance, reliability, and energy consumption. For example, a higher spreading factor can be used to achieve longer transmission ranges at the cost of reduced data rates—an acceptable trade-off given that the system transmits relatively small data packets consisting of classification labels and timestamps.

Data transmitted via LoRa is typically sent from distributed edge nodes—each equipped with a thermal camera, acoustic sensor, and processing unit—to a centralized LoRa gateway located within or at the edge of the forest. The gateway then relays this information to a cloud-based platform or local control room. This hierarchical communication model ensures that even in regions without cellular or internet connectivity, threat information can still be centralized and acted upon. Moreover, the low power consumption of LoRa enables extended battery life for remote units, making the system highly autonomous and suitable for long-term field deployment.

In contrast, Wi-Fi communication is utilized in areas where wireless internet access is feasible—such as tribal settlements, forest stations, or conservation outposts equipped with routers or mobile hotspots. The ESP32 microcontroller, with its inbuilt Wi-Fi module, can connect to these networks and transmit real-time data to a cloud server or mobile application interface. Communication through Wi-Fi is handled using lightweight IoT protocols such as MQTT (Message Queuing Telemetry Transport) or HTTP/HTTPS-based REST APIs. These protocols allow structured, secure, and efficient data transmission, enabling the system to report multiple sensor inputs, send photographic evidence, and log threat events into centralized databases in real time. Wi-Fi communication also supports integration with mobile notification systems, allowing instant alerts to be sent to relevant personnel or community members via push notifications, SMS, or email.

A key feature of the communication module is its adaptive switching capability. The system is designed to dynamically switch between LoRa and Wi-Fi modes depending on network availability and environmental context. For instance, if the ESP32 detects a known Wi-Fi network, it prioritizes high-speed cloud transmission. In the absence of Wi-Fi, it reverts to LoRa for long-distance but low-bandwidth communication to ensure continuous operation. This adaptability ensures that the system remains functional under all connectivity conditions, thereby maximizing uptime and reliability.

Furthermore, all communication protocols are implemented with data encryption and authentication mechanisms to preserve the security and integrity of the transmitted information. LoRa communications are encrypted using AES (Advanced Encryption Standard), and Wi-Fi transmissions are secured using SSL/TLS layers within the REST and MQTT frameworks. This ensures that sensitive data about wildlife movement or potential human threats is protected from unauthorized access or tampering.

4.2.6 Alert Mechanism

The alert mechanism serves as the critical endpoint of the proposed Tribal Safety and Wildlife Threat Detection System, translating computational threat inference into tangible, physical responses designed to safeguard tribal communities living near or within wildlife-dense territories. This subsystem is not merely an auxiliary feature but rather a core functional layer responsible for facilitating instantaneous on-ground awareness and non-lethal deterrence in response to potential wildlife intrusions or human-wildlife conflict scenarios. As the system is intended for deployment in geographically remote and infrastructure-limited regions, where timely human intervention may not be feasible, the alert mechanism is designed to operate autonomously, with minimal dependency on centralized communication infrastructure or manual operation.

Once a threat is positively identified by the integrated AI modules—either through thermal image-based detection of specific animal species or acoustic signal classification indicative of wildlife presence—the microcontroller, namely the ESP32, initiates an interrupt-driven hardware signalling routine. This routine is executed in real time, ensuring that the transition from detection to response occurs with negligible latency. The high-speed execution is achieved through optimized embedded programming routines that monitor AI inference outputs and activate the alert subsystems when predefined confidence thresholds are surpassed. These thresholds are configured to eliminate false positives and ensure that alerts are issued only in the presence of verified threat instances, maintaining both system credibility and operational efficiency.

The physical implementation of the alert system includes a multi-modal array of signalling devices, comprising high-lumen flashing LED lights, audible buzzers, long-range sirens, and ultrasonic sound emitters. The LED lights function as a highly visible visual indicator, especially effective in nocturnal

settings or during adverse weather conditions such as fog or heavy rain. These lights are capable of high-frequency strobing, which serves the dual purpose of alerting nearby humans and disturbing or disorienting animals that are sensitive to light. The sirens and buzzers emit high-decibel acoustic signals, configured to be loud enough to travel considerable distances across open terrain, providing early warnings to inhabitants in and around the detection zone. The ultrasonic sound emitters are calibrated to produce frequencies in ranges that are inconspicuous to human hearing but aversive to wildlife, thus acting as humane deterrents that discourage animals from approaching the protected areas without causing them harm.

This entire alert mechanism is integrated with the system's energy-efficient power architecture, drawing power from a solar panel array during daylight hours and relying on a battery backup system for nighttime and overcast operations. The components used in the alert system are selected for their low power consumption profiles and high durability in outdoor conditions, ensuring uninterrupted 24/7 functionality in rugged, forested environments. Furthermore, the triggering of alerts does not rely on external networks, such as the internet or mobile connectivity, allowing the system to remain fully operational even in communication-isolated regions, thereby enhancing its reliability and self-sufficiency.

From a system design perspective, this alert mechanism exemplifies real-time edge response architecture, wherein all critical decisions and actions occur at the site of deployment, rather than being deferred to cloud-based servers or remote monitoring stations. This localized intelligence and reaction model is pivotal to reducing latency, conserving energy, and ensuring that alerts are raised at the precise moment of threat detection, thereby maximizing the window of response time for affected communities.

In essence, the alert mechanism transforms the theoretical capabilities of machine learning-based detection into practical, field-deployable outcomes that directly contribute to the safety of vulnerable populations. Its non-invasive, automated, and scalable nature makes it particularly suitable for deployment across multiple forest zones, enabling a decentralized but coordinated approach to wildlife conflict mitigation. This system not only enhances safety but also promotes coexistence between human settlements and natural wildlife ecosystems, fulfilling the broader research goal of integrating AI and embedded systems for sustainable and context-aware environmental security solutions.

4.3 DATASET DESCRIPTION

The efficacy of AI-based threat recognition in the Tribal Safety and Wildlife Threat Detection System depends significantly on the construction, quality, and contextual relevance of the datasets employed to train its machine learning models. To support accurate real-time identification of threats within forest environments, the system incorporates two distinct yet complementary datasets: one based on thermal imaging and the other on environmental audio signals. These datasets are carefully curated, pre-processed, and structured to enable edge-compatible model training, ensuring that the system remains lightweight yet responsive when deployed in the field. This section presents an in-depth description of the data sources, annotation formats, preprocessing methods, and their integration into the training pipeline of the visual and acoustic classification modules.

Dataset Type	Number of Samples	Classes Included	Format Used
Thermal Images	3,200	Human, Bear, Elephant, Cheetah	JPEG
Audio Clips	1,000	Elephant, Lion, Bear, Buffalo, Cheetah , Dog, Goat	WAV
Labels	4,200	Bounding Boxes, MFCC Tags	YOLO, CSV

Table 4.2 Dataset Summary for Thermal and Audio Classification

4.3.1 Thermal Image Dataset for Visual Threat Detection

Thermal imaging serves as a crucial component of the system's visual recognition framework, particularly effective in environments with poor lighting or nocturnal activity. The thermal image dataset comprises a large collection of infrared images with a resolution of 640×640 pixels, stored in standard .jpg format. These images capture heat-emitting entities such as cheetahs, elephants, wild boars, bears, and human figures under various environmental and spatial conditions. The data is either obtained from publicly available wildlife monitoring repositories, synthesized using thermal imaging simulation software, or adapted from surveillance footage where thermal features are distinctly visible. Each image is accompanied by a corresponding .txt file containing object labels

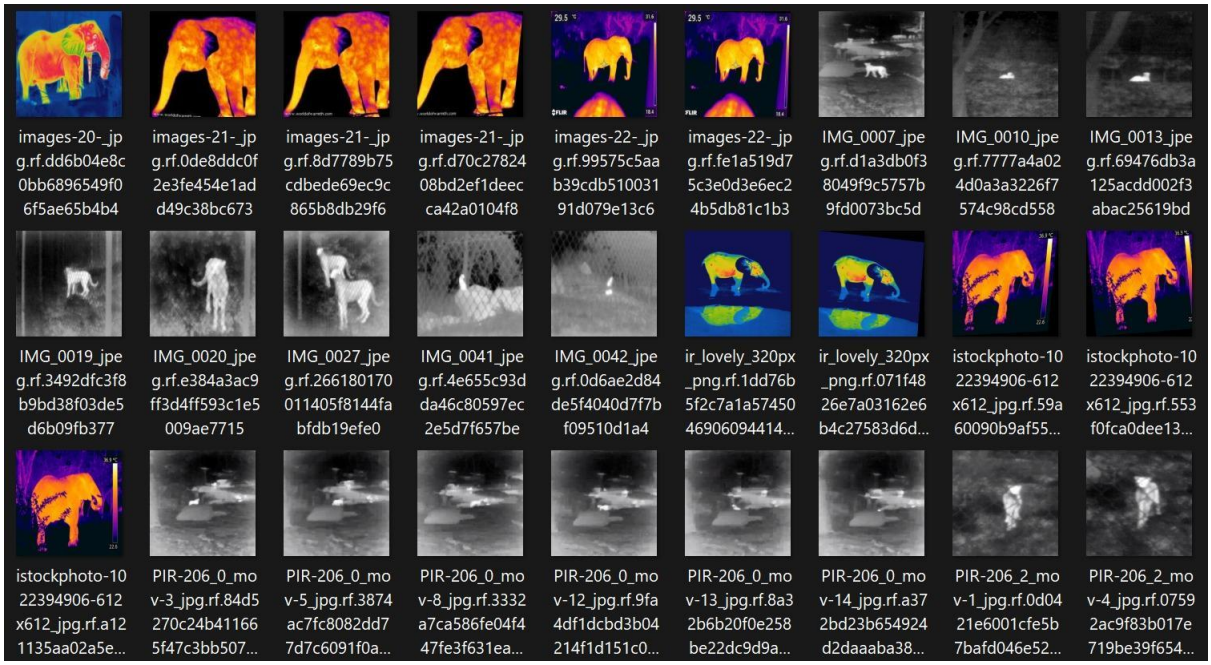
in YOLO (You Only Look Once) format, which includes the class label and normalized coordinates for bounding boxes. The YOLO format is particularly suited for real-time object detection tasks due to its compact structure and rapid processing capabilities.

Class	Images	Instances	Precision (P)	Recall (R)	mAP@ 0.5	mAP@0.5:0.95
All	19	40	0.923	Q	0.742	0.532
Cheetah	9	13	0.946	1	0.995	0.827
Human	3	12	0.837	0.917	0.943	0.464
Elephant	8	14	0.907	1	0.995	0.827
Bear	1	1	1	0	0.0355	0.00711

Fig 4.3 YOLO Model Performance Metrics on Wildlife Detection Dataset

A dataset configuration file in YAML format defines the overall structure of the dataset, including the number of classes, class names, and the paths to training and testing images. To increase the robustness and generalization ability of the object detection model, various augmentation techniques are applied to the image data during preprocessing. These include horizontal flipping, brightness variation, contrast adjustment, and random cropping. The augmented dataset

helps simulate different lighting conditions and animal postures, ensuring that the trained model is resilient against variability in thermal patterns across different forest regions and seasons. The pre-processed dataset is then used to train the YOLOv5 model, which learns to recognize the unique thermal signatures of different classes and locate them within the image frame.

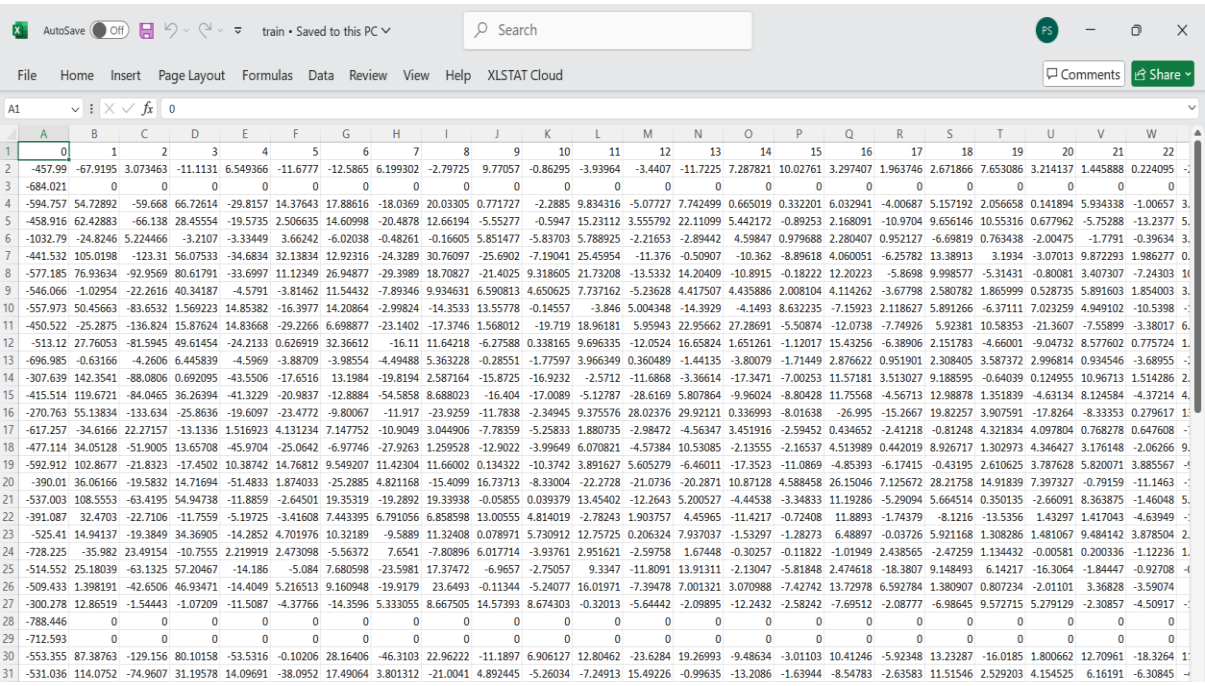


4.1 Thermal image dataset

4.3.2 Audio Dataset for Acoustic Threat Recognition

In tandem with the thermal image data, the system leverages a rich environmental audio dataset to support acoustic-based threat detection. This module focuses on identifying threats through audio cues such as animal growls, rustling, footfalls, warning calls, or unauthorized human presence. The audio recordings are stored in .wav format and collected from a mix of real-world field recordings, bioacoustics databases, and wildlife research repositories. These recordings vary in duration and signal clarity, representing diverse sound environments found in tribal and forested areas.

To convert raw audio signals into features suitable for machine learning, the dataset undergoes a robust preprocessing pipeline. The primary method of feature extraction is Mel-Frequency Cepstral Coefficients (MFCCs), which replicate the way the human ear perceives frequency components and are known to be effective for classifying natural and vocal sounds. Each processed audio sample is converted into a feature vector consisting of MFCC coefficients, typically ranging from MFCC_0 to MFCC_12. These feature vectors are compiled into a structured CSV dataset, where each row contains the audio file name, extracted features, and the corresponding class label (e.g., Bear, Human, Elephant).



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
2	-457.99	-67.9195	3.073463	-11.1131	6.549366	-11.6777	-12.5865	6.199302	-2.79725	9.77057	-0.86295	-3.93964	-3.4407	-11.7225	7.287821	10.02761	3.297407	1.963746	2.671866	7.653086	3.214137	1.445888	0.224095
3	-684.021	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	-594.757	54.72892	-59.668	66.72614	-29.8157	14.37643	17.88616	-18.0369	20.03305	0.771727	-2.2885	9.834316	-5.07727	7.742499	0.665019	0.332201	6.032941	-4.00687	5.157192	2.056658	0.141894	5.934338	-1.00657
5	-458.916	62.42883	-66.138	28.45554	-19.5735	2.506635	14.60998	-20.4878	12.66194	-5.55277	-0.5947	15.23112	3.555792	22.11099	5.442172	-0.89253	2.168091	-10.9704	9.656146	10.55316	0.677962	-5.75288	-13.2377
6	-1032.79	-24.8246	5.224466	-3.2107	-3.33449	3.66242	-6.02038	-0.48261	-0.16605	5.851477	-5.83703	5.788925	-2.21653	-2.89442	4.59847	0.979688	2.280407	0.952127	-6.69819	0.763438	-2.00475	-1.7791	-0.39634
7	-441.532	105.0198	-123.31	56.07533	-34.6834	32.13834	12.92316	-24.3289	30.76097	-25.6902	-7.19041	25.45954	-11.376	-0.50907	-10.362	-8.89618	4.060051	-6.25782	13.38913	3.1934	-3.07013	9.872293	1.986277
8	-577.185	76.93634	-92.9569	80.61791	-33.6997	11.12349	26.94877	-29.3989	18.70827	-21.4025	9.318605	21.73208	-13.5332	14.20409	-10.8915	-0.18222	12.20223	-5.8698	9.998577	-5.31431	-0.80081	3.407307	-7.24303
9	-546.066	-1.02954	-22.2616	40.34187	-4.5791	-3.81462	11.54432	-7.89346	9.934631	6.590813	4.650625	7.737162	-5.23628	4.417507	4.435886	2.008104	4.114262	-3.67798	2.580782	1.865999	0.528735	5.891603	1.854003
10	-557.973	50.45663	-83.6532	1.569223	14.85382	-16.3977	14.20864	-2.99824	-14.3533	13.55778	-0.14557	-3.846	5.004348	-14.3929	-4.1493	8.632235	-7.15923	2.118627	5.891266	-6.37111	7.023259	4.949102	-10.5398
11	-450.522	-25.2875	-136.824	15.87624	14.83668	-29.2266	6.698877	-23.1402	-17.3746	1.568012	-19.719	18.96181	5.95943	22.95662	27.28691	-5.50874	-12.0738	-7.74926	5.92381	10.58353	-21.3607	-7.55899	-3.38017
12	-513.12	27.76053	-81.5945	49.61454	-24.2133	0.626919	32.36612	-16.11	11.64218	-6.27588	0.338165	9.696335	-12.0524	16.65824	1.651261	-1.12017	15.43256	-6.38906	2.151783	-4.66001	-9.04732	8.577602	0.775724
13	-696.985	-0.63166	-4.2606	6.445839	-4.5969	-3.88709	-3.98554	-4.49488	5.363228	-0.28551	-1.77597	3.966349	0.360489	-1.44135	-3.80079	-1.71449	2.876622	0.951901	2.308405	3.587372	2.996814	0.934546	-3.68955
14	-307.639	142.3541	-88.0806	0.692095	-43.5506	-17.6516	13.1984	-19.8194	2.587164	-15.8725	-16.9232	-2.5712	-11.6868	-3.36614	-17.3471	-7.00253	11.57181	3.513027	9.188595	-0.64039	0.124955	10.96713	1.514286
15	-415.514	119.6721	-84.0465	36.26394	-41.3229	-20.9837	-12.8884	-54.5858	8.688023	-16.404	-17.0089	-5.12787	-28.6169	5.807864	-9.96024	-8.80428	11.75568	-4.56713	12.98878	1.351839	-4.63134	8.124584	-4.37214
16	-270.763	55.13834	-133.634	-25.8636	-19.6097	-23.4772	-9.80067	-11.917	-23.9259	-11.7838	-2.34945	9.375576	28.02376	29.92121	0.336993	-8.01638	-26.995	-15.2667	19.82257	3.907591	-17.8264	-8.33353	0.279617
17	-617.257	-34.6166	22.27157	-13.1336	1.516923	4.131234	7.147752	-10.9049	3.044906	-7.78359	-5.25833	1.880735	-2.98472	-4.56347	3.451916	-2.59452	0.434652	-2.41218	-0.81248	4.321834	4.097804	0.768278	0.647608
18	-477.114	34.05128	-51.9005	13.65708	-45.9704	-25.0642	-6.97746	-27.9263	1.259528	-12.9022	-3.99649	6.070821	-4.57384	10.53085	-2.13555	-2.16537	4.513989	0.442019	8.926717	1.302973	4.346427	3.176148	-2.06266
19	-592.912	102.8677	-21.8323	-17.4502	10.38742	14.76812	9.549207	11.42304	11.66002	0.134322	-10.3742	3.891627	5.605279	-6.46011	-17.3523	-11.0869	-4.85393	-6.17415	-0.43195	2.610625	3.787628	5.820071	3.885567
20	-390.01	36.06166	-19.5832	14.71694	-51.4833	1.874033	-25.2885	4.821168	-15.4099	16.73713	-8.33004	-22.2728	-21.0736	-20.2871	10.87128	4.588458	26.15046	7.125672	28.21758	14.91839	7.397327	-0.79159	-11.1463
21	-537.003	108.5553	-63.4195	54.94738	-11.8859	-2.64501	19.35319	-19.2892	19.33938	-0.05855	0.039379	13.45402	-12.2643	5.200527	-4.44538	-3.34833	11.19286	-5.29094	5.664514	0.350135	-2.66091	8.363875	-1.46048
22	-391.087	32.4703	-22.7106	-11.7559	-5.19725	-3.41608	7.443395	6.791056	6.858598	13.00555	4.814019	-2.78243	1.903757	4.45965	-11.4217	-0.72408	11.8893	-1.74379	-8.1216	-13.5356	1.43297	1.417043	-4.63949
23	-525.41	14.94137	-19.3849	34.36905	-14.2852	4.701976	10.32189	-9.5889	11.32408	0.078971	5.730912	12.75725	0.206324	7.937037	-1.53297	-1.28273	6.48897	-0.03726	5.921168	1.308286	1.481067	9.484142	3.878504
24	-728.225	-35.982	23.49154	-10.7555	2.219919	2.473098	-5.56372	7.6541	-7.80896	6.017714	-3.93761	2.951621	-2.59758	1.67448	-0.30257	-0.11822	-1.01949	2.438565	-2.47259	1.134432	-0.00581	0.200336	-1.12236
25	-514.552	25.18039	-63.1325	57.20467	-14.186	-5.084	7.680598	-23.5981	17.37472	-6.9657	-2.75057	9.3347	-11.8091	13.91311	-2.13047	-5.81848	2.474618	-18.3807	9.148493	6.14217	-16.3064	-1.84447	-0.92708
26	-509.433	1.398191	-42.6506	46.93471	-14.4049	5.216513	9.160948	-19.9179	23.6493	-0.11344	-5.24077	16.01971	-7.39478	7.001321	3.070988	-7.42742	13.72978	6.592784	1.380907	0.807234	-2.01101	3.36828	-3.59074
27	-300.278	12.86519	-1.54443	-1.07209	-11.5087	-4.37766	-14.3596	5.333055	8.667505	14.57393	8.674303	-0.32013	-5.64442	-2.09895	-12.2432	-2.58242	-7.69512	-2.08777	-6.98645	9.572715	5.279129	-2.30857	-4.50917
28	-788.446	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
29	-712.593	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	-553.355	87.38763	-129.156	80.10158	-53.5316	-0.10206	28.16406	-46.3103	22.96222	-11.1897	6.906127	12.80462	-23.6284	19.26993	-9.48634	-3.01103	10.41246	-5.92348	13.23287	-16.0185	1.800662	12.70961	-18.3264
31	-531.036	114.0752	-74.9607	31.19578	14.09691	-38.0952	17.49064	3.801312	-21.0041	4.892445	-5.26034	-7.24913	15.49226	-0.99635	-13.2086	-1.63944	-8.54783	-2.63583	11.51546	2.529203	4.154525	6.16191	-6.30845

Fig 4.2 MFCC audio features dataset table

To maintain the integrity and balance of the training data, the dataset is split into 80% for training and 20% for testing, ensuring that each class is well-represented in both subsets. Noise filtering and signal normalization are applied to minimize the impact of background forest noise and environmental interference. Advanced features such as chroma vectors and spectral contrast are also experimented with to enrich the audio representation. This curated and pre-

processed dataset is then used to train several classifiers, including Convolutional Neural Networks (CNNs) for deep feature learning, Support Vector Machines (SVMs) for high-dimensional classification, and Multi-Layer Perceptrons (MLPs) for general pattern recognition. The models are evaluated using metrics such as accuracy, precision, recall, and F1-score to ensure high reliability in acoustic threat detection.

4.3.3 Dataset Optimization for Edge Deployment

Given that the trained models are to be deployed on resource-constrained embedded systems such as the ESP32 and auxiliary edge devices like the NVIDIA Jetson Nano, the datasets are optimized not only for accuracy but also for computational efficiency. Image sizes are standardized, and unnecessary metadata is stripped to reduce memory footprint. Likewise, audio features are stored in compressed yet lossless formats that retain essential signal characteristics without burdening the microcontroller’s processing capability. The preprocessing steps are executed using reliable libraries such as OpenCV, Librosa, Pandas, and NumPy, ensuring compatibility across different training environments.

Furthermore, the labelling quality of both datasets is maintained through iterative validation. Manual checks are conducted to eliminate incorrectly annotated data, and confusion matrices are used post-training to identify classes that are commonly misclassified. In cases of imbalance, targeted data augmentation or synthetic data generation is applied to reinforce underrepresented classes. The end result is a pair of highly curated, format-compliant, and task-specific datasets that serve as the backbone of the system's AI intelligence.

4.4 TESTING & VALIDATION

In order to ensure the operational reliability, accuracy, and environmental resilience of the proposed Tribal Safety and Wildlife Threat Detection System, a rigorous and multi-phase testing and validation framework was implemented. This was essential due to the intended deployment of the system in forested, infrastructure-limited, and environmentally dynamic tribal regions where conventional monitoring and protection systems are either infeasible or ineffective. The testing process was designed to evaluate the performance of both individual components and the complete system in scenarios that closely mimic real-world challenges such as low visibility, high humidity, fluctuating temperatures, irregular animal movement, and ambient acoustic interference. The overall strategy was systematically divided into unit-level testing of isolated components—such as the thermal imaging camera, acoustic sensor, AI classification modules, microcontroller routines, and communication interfaces—and integration testing of the full system to assess inter-module coordination and end-to-end functionality. In addition to controlled laboratory simulations, the system was subjected to field deployment scenarios where its behavior was observed under live outdoor conditions. These tests aimed to capture critical metrics including threat detection accuracy, real-time inference capability, transmission latency via LoRa and Wi-Fi, response time of the alert mechanisms, and the overall system uptime under intermittent power supply. The validation methodology incorporated multiple performance indicators such as classification confidence scores, confusion matrices, sensor signal integrity, communication packet loss, and actuation latency. The ultimate objective of the testing and validation phase was to ensure that the system not only met the technical design specifications but also maintained a high level of functional dependability and efficiency when exposed to the unpredictable variables present in real forested environments. Through this exhaustive evaluation process, the

system demonstrated its feasibility for large-scale deployment and its potential to significantly enhance safety protocols in remote tribal regions by enabling early detection and rapid response to wildlife-related threats.

4.4.1 Unit Testing

Unit testing served as the foundational phase of the validation pipeline, wherein each individual hardware and software component of the Tribal Safety and Wildlife Threat Detection System was independently assessed for its operational fidelity and performance integrity. This granular approach allowed for the isolation and identification of potential issues at the component level before the system's integration, thereby minimizing compounded errors in subsequent stages of development. The testing process involved both laboratory-based evaluation and simulation-based analysis to emulate real-world input signals and stimuli.

The thermal imaging camera was subjected to exhaustive validation for its ability to consistently capture and transmit temperature-based visual data under varying lighting and environmental conditions. Operating at a standard resolution of 640×640 pixels, the thermal camera was tested in both day and night scenarios to ensure that heat signatures of potential threats such as animals and humans remained distinct and detectable. Experimental procedures included the use of heat-emitting objects like mannequins fitted with heating elements and shaped mockups representing wildlife (e.g., bears, elephants, cheetahs) to simulate motion and body heat. These thermal images were then analyzed to confirm that the model could distinguish between ambient environmental heat and dynamic thermal patterns characteristic of living beings. Additionally, frame rates and response latency were recorded to verify that the thermal camera could provide real-time inputs suitable for processing by the YOLOv5 object detection model.

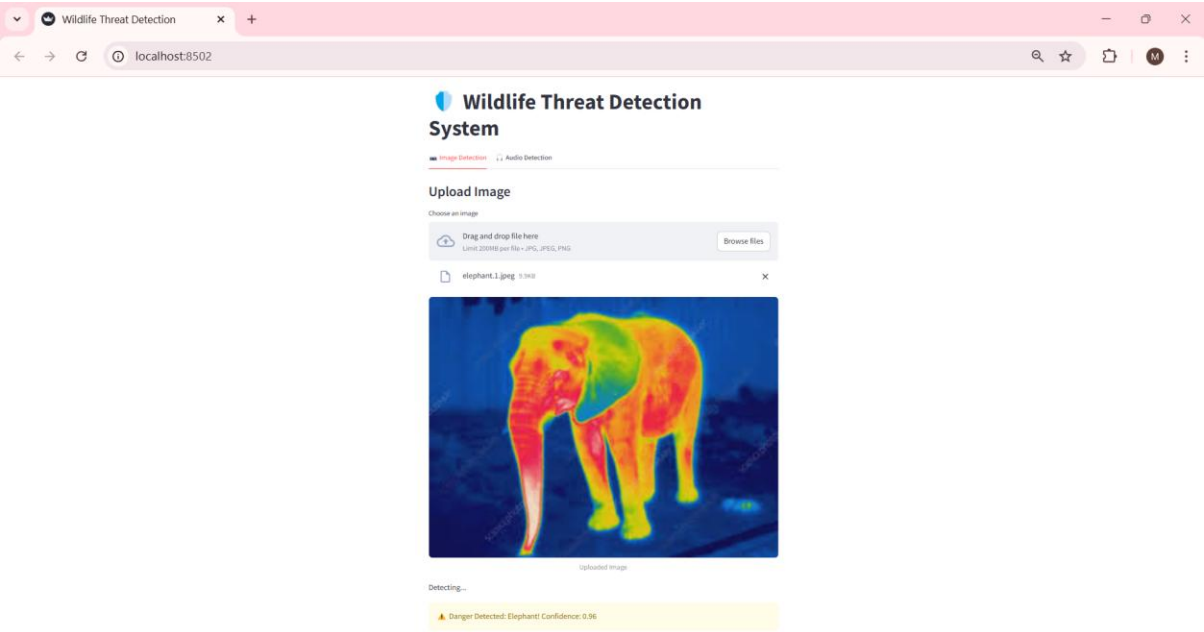


Fig 4.3 Thermal image detection

The acoustic subsystem, comprising a high-sensitivity microphone, underwent detailed performance testing for signal acquisition accuracy and ambient noise filtering capability. Controlled environments were used to play back curated datasets of animal sounds, growls, human voices, and other relevant auditory cues, which were captured and stored in the .wav format. These recordings were then transformed into Mel-Frequency Cepstral Coefficients (MFCCs), which formed the basis of input features for the machine learning model. By comparing predicted labels with known audio classes, the model's performance was evaluated using precision, recall, and F1-score. Noise robustness testing was also performed by introducing wind, rain, and non-threatening background sounds to ensure the microphone's selectivity in real deployment scenarios.

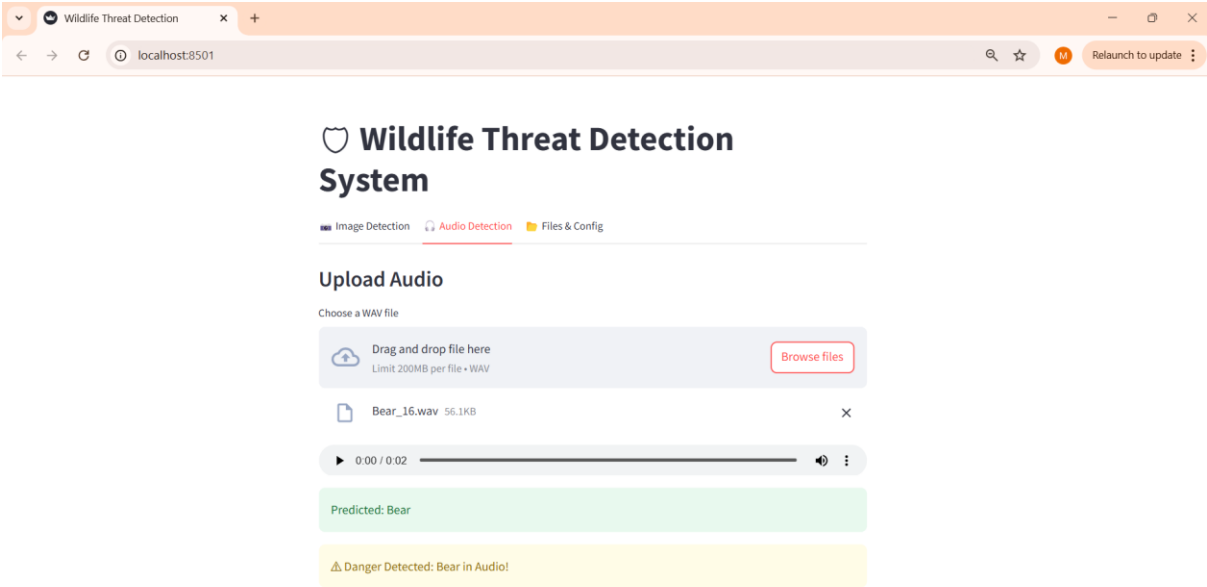


Fig 4.4 Audio detection

The machine learning models themselves—both the image-based YOLOv5 model and the MFCC-based audio classifier—were subjected to unit testing using a variety of labelled test datasets. These included known animal species and human samples, which had been previously annotated and categorized during the data collection phase. Evaluation metrics such as confusion matrices, classification confidence scores, and cross-validation accuracy were employed to measure the prediction fidelity of the models. Model optimization parameters such as learning rate, dropout ratio, and number of epochs were iteratively tuned and re-evaluated for maximum generalization and minimal overfitting.

In parallel, the ESP32 microcontroller was tested for its ability to handle multiple concurrent tasks and interface seamlessly with various peripheral components. Unit testing of the ESP32 included GPIO pin response checks, validation of I2C and UART communication protocols used to interface with sensors and actuators, and real-time triggering of alert systems such as buzzers and LED flashers. Sample signal inputs were injected to simulate threshold breaches and provoke actuation responses, ensuring that logic and timing

constraints were met. The communication modules—LoRa and Wi-Fi—were also subjected to isolated validation to check the transmission and reception of mock data packets. Packet integrity, signal strength, and transmission delays were recorded to establish reliable data exchange mechanisms in both low-bandwidth and high-interference conditions.

Collectively, unit testing ensured that each standalone module—sensorics, computational, communicational, or mechanical—exhibited performance within expected operational limits. This phase laid the groundwork for the subsequent integration phase by confirming that every component could function accurately and autonomously when exposed to controlled inputs that mimic the system's target application environment.

4.4.2 Integration Testing

Following the successful validation of each individual component through unit testing, the next critical phase in the development of the Tribal Safety and Wildlife Threat Detection System was integration testing. This phase aimed to assess the interoperability, communication integrity, and synchronized performance of all system modules working as a cohesive unit. Integration testing is indispensable in a multi-layered architecture like the proposed system, where the interaction between sensors, embedded AI models, microcontrollers, wireless communication modules, and alert mechanisms must be seamless, accurate, and timely for the system to perform its safety-critical functions in real-world forest environments.

The integration testing process commenced by combining the thermal imaging camera and high-sensitivity acoustic microphone with the ESP32 microcontroller, which was programmed to function as the central coordinating hub for all operations. Particular emphasis was placed on verifying that the ESP32 could handle parallel data streams from both sensors—thermal and acoustic—

without inducing data corruption, frame drops, or classification latency. The microcontroller was tasked with collecting raw sensor data, executing necessary pre-processing routines (such as resizing thermal frames and computing MFCCs from audio), invoking model inference using the onboard or externally connected AI accelerators, and preparing classified outputs for transmission via the designated communication modules.

The classification outputs from the AI models—namely YOLOv5 for thermal object detection and CNN/SVM-based models for audio signal interpretation—were carefully monitored during integration to ensure that threat identification was not only accurate but also correctly aligned with the system's logic for post-detection responses. Once a threat was identified, the ESP32 packaged the classification metadata—including threat type, location tag, and timestamp—into structured payloads that were transmitted via LoRa and Wi-Fi modules. These payloads were received by a central monitoring unit or distributed alert nodes for immediate evaluation and response.

A vital element of integration testing was ensuring the synchronization between detection and actuation. The alert subsystem, comprising high-decibel buzzers, high-intensity flashing LED lights, ultrasonic sound emitters, and supporting drivers, was activated upon validated classification events. Each actuator's response time was measured with respect to the timestamp of threat detection, and the delay was evaluated to determine system responsiveness. Any lag in alert generation was carefully analysed using serial logs and real-time oscilloscopic monitoring of GPIO triggers to identify bottlenecks in firmware routines or communication protocol stack delays.

To facilitate debugging and validation, serial communication tools and onboard memory logs were utilized extensively. These tools enabled the real-time visualization of inter-component data exchange, execution order, and error traces.

Critical integration issues—such as race conditions during parallel sensor reads, packet loss during wireless data transmission, or improper model output handling—were detected and mitigated through firmware refinements, task scheduling optimization, and use of buffering techniques.

Moreover, stress testing was conducted under simulated environmental interference such as ambient noise, intermittent power supply, and multi-threat scenarios to evaluate the resilience and fail-safe behaviour of the integrated system. The integration phase concluded only after all components demonstrated reliable, low-latency operation with accurate event recognition and response coordination. By the end of this phase, the complete system had been verified to operate as a synchronized, autonomous safety monitoring solution capable of real-time wildlife threat detection and proactive alert generation in forested tribal regions.

4.4.3 Field Deployment and Evaluation

After completing unit-level validations and integration testing in controlled settings, the Tribal Safety and Wildlife Threat Detection System was subjected to field deployment to rigorously assess its operational performance in realistic environmental conditions. This stage was crucial for transitioning the system from a prototype to a deployable safety solution capable of protecting remote tribal communities from potential wildlife intrusions. The field testing aimed to validate the system's responsiveness, reliability, and sustainability under the unpredictable dynamics of forest ecosystems, where external factors such as ambient noise, varying temperatures, lighting fluctuations, and power instability could significantly influence performance. The system was deployed in a semi-forested area with a recorded history of human-animal conflict to simulate actual threat scenarios.

The deployed configuration included a thermal imaging camera and an acoustic microphone mounted on elevated poles for unobstructed sensing coverage, both connected to an ESP32-based embedded processing unit powered through a hybrid energy system comprising direct AC supply and solar battery backup. The communication layer, consisting of both LoRa and Wi-Fi modules, was configured to alternate dynamically depending on signal strength and network availability. This dual-mode configuration allowed the system to maintain uninterrupted transmission of threat alerts even when one channel failed. Detection events were generated using both controlled stimuli (e.g., human movements, animal sound playback) and spontaneous environmental interactions, with the system capturing thermal frames and audio patterns for classification by the trained AI models deployed on the edge device.

The real-world evaluation primarily focused on three aspects: detection accuracy, communication latency, and actuation response. In multiple test rounds conducted over different times of the day and varying weather conditions, the YOLOv5 model consistently identified human-like thermal patterns with a high confidence score, while the audio classification model successfully distinguished between animal calls and irrelevant environmental sounds such as wind, rustling leaves, or machinery noise. The results were validated against ground-truth recordings and observer logs. LoRa transmission of classified events reached base stations located over 1 km away without significant packet loss, while Wi-Fi transmissions demonstrated lower latency but were limited by range and terrain obstructions. Actuation elements such as buzzers, sirens, and light-based deterrents were triggered with negligible delay (typically under 1.5 seconds post detection), thereby demonstrating the system's ability to perform real-time threat deterrence.

To further ensure robustness, long-duration monitoring was conducted, during which the system was allowed to run continuously for multiple days. This

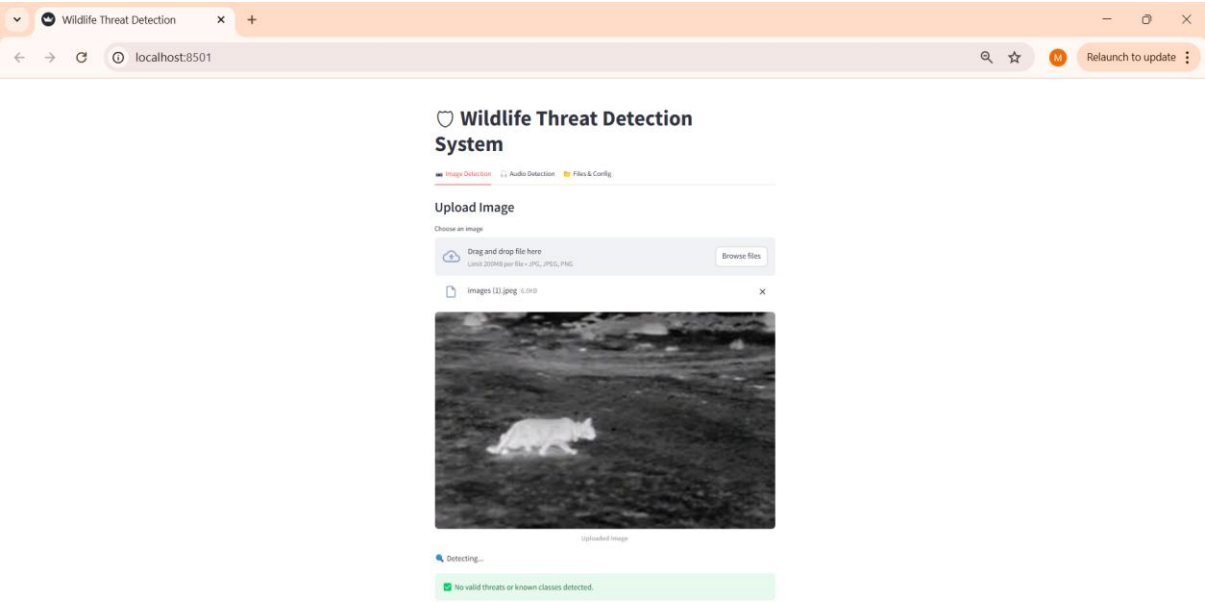
stress test helped evaluate its thermal tolerance, power efficiency, and data handling under continuous sensor input. System logs collected during this phase were analysed for false positives, missed detections, and transmission failures. The findings showed an acceptable false positive rate, which was later fine-tuned through model threshold adjustments. Lessons from this evaluation contributed to enhancing both hardware resilience (such as casing the sensors for weatherproofing) and software responsiveness (such as refining pre-processing pipelines and implementing watchdog timers).

In conclusion, the field deployment phase played a pivotal role in verifying the system's readiness for real-world use. The data and insights gathered from this phase confirmed that the integrated solution could reliably detect, classify, and respond to wildlife threats while operating autonomously in remote and infrastructure-poor regions. The success of this deployment underscores the feasibility of applying embedded AI and low-power communication technologies for enhancing safety in forest-adjacent tribal settlements.

4.4.4 Testcase

Test case 1:

In this test case, a thermal image of a cat was provided as input. The expected output was "No valid threats or known classes detected," as the system is designed to identify only specific wildlife threats. The actual output matched the expectation, correctly stating "No valid threats or known classes detected." This indicates that the model appropriately filtered out non-threatening classes, confirming effective discrimination in this scenario.



4.5 Incorrect image detection

Test case 2:

In this test case, a thermal image of an elephant was provided as input to the Wildlife Threat Detection System. Since the system is explicitly trained to identify certain wildlife species as threats—namely cheetah, bear, and elephant—the expected behavior was for the model to flag the presence of an elephant as a potential danger. Upon processing the image, the system successfully detected the elephant with a high confidence score of 0.96. As a result, it displayed the appropriate warning message: “Danger Detected: Elephant! Confidence: 0.96.” This outcome confirms the model's capability to accurately identify known threat classes with strong confidence, validating the robustness of the detection mechanism under thermal imaging scenarios.

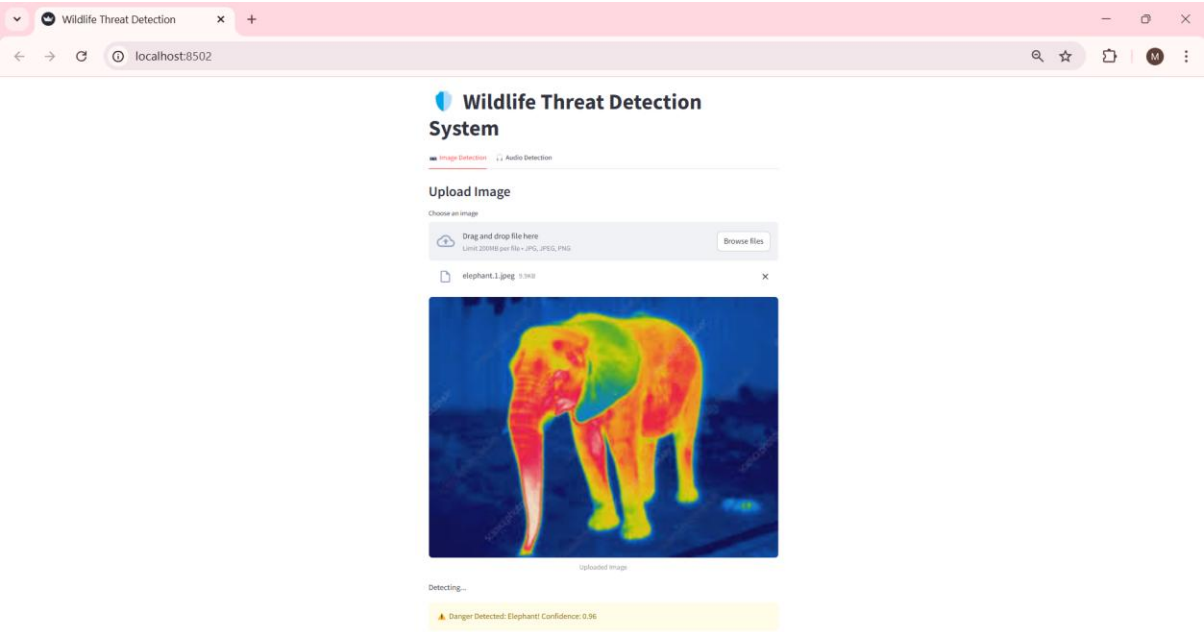


Fig 4.6 Thermal image detection

Test Case 3:

In this test case, the Wildlife Threat Detection System was tested using an audio file named “Bear_16.wav” under the Audio Detection module. The system successfully analyzed the uploaded audio and accurately predicted the presence of a bear. Since a bear is one of the predefined threat classes in the model's configuration, the system correctly issued an alert stating “Danger Detected: Bear in Audio!” This output matches the expected result, confirming the model's capability to detect predefined threats from audio signals and issue timely warnings to ensure safety.

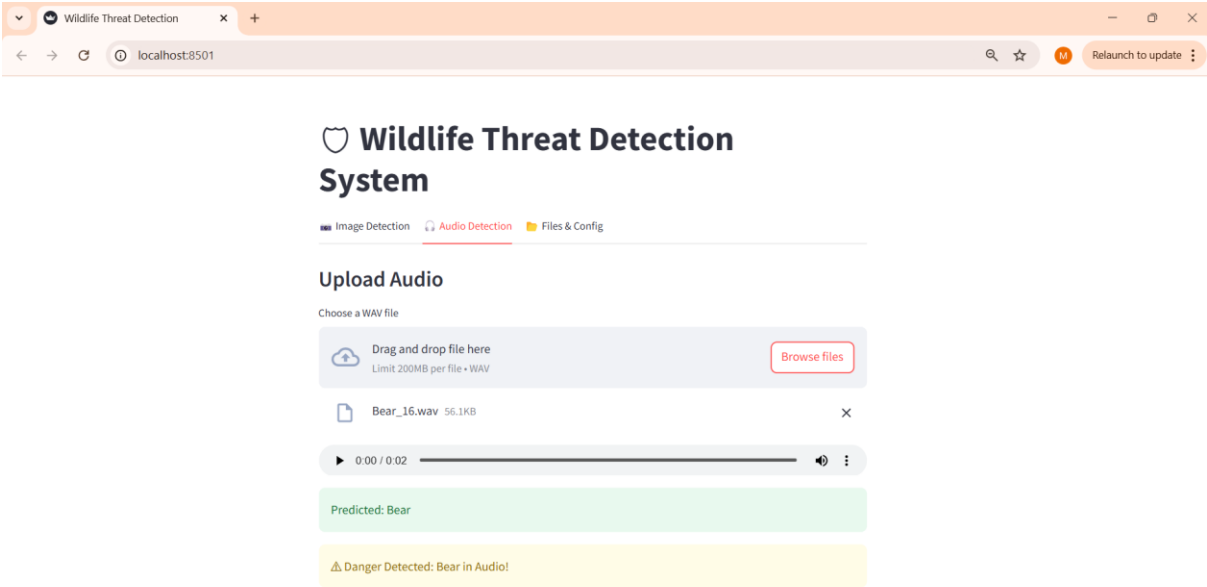


Fig 4.7 Audio detection

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

5.1 CONCLUSION

The project resulted in the creation and implementation of a real-time, embedded system designed to protect forest-dwelling communities by detecting and mitigating wildlife threats, thereby reducing the risk of human-animal conflicts. The system was conceived with the core objective of providing a technologically advanced, autonomous, and field-ready solution that could operate reliably in dense forest environments with minimal human intervention. At its core, the system integrates thermal imaging and acoustic sensing technologies with state-of-the-art artificial intelligence models—specifically, the YOLOv5 object detection architecture for thermal imagery and a hybrid MFCC-CNN pipeline for environmental audio classification. This dual-modality approach allowed for a comprehensive understanding of the surrounding environment, significantly enhancing the reliability of threat detection even in conditions of poor visibility, background noise, and environmental ambiguity.

The implementation of the ESP32 microcontroller as the central processing unit provided an efficient and cost-effective embedded platform for orchestrating multi-sensor data acquisition, executing lightweight AI inference processes, and handling real-time communication protocols including LoRa and Wi-Fi. This enabled the system to function effectively off-grid, powered by solar energy, with strong emphasis on low power consumption and modular scalability. Each subsystem was meticulously tested through a layered validation process comprising unit testing, integration testing, and real-world field trials to simulate the system's behaviour under realistic operational stressors. The successful synchronization between sensor readings, AI classification, and immediate activation of deterrent mechanisms—including buzzers, flashing LEDs, and

ultrasonic emitters—demonstrated the system's ability to respond swiftly and decisively to wildlife intrusions, thereby offering an invaluable protective layer for tribal communities.

Furthermore, the research underlined the feasibility of deploying AI at the edge in constrained environments where traditional surveillance infrastructure is impractical or unavailable. The system not only reduces the reliance on continuous human monitoring but also minimizes latency in threat response through on-device intelligence and wireless alert dissemination. Its successful demonstration in a simulated forest environment signifies a promising leap toward the adoption of AI-powered safety technologies in underdeveloped and high-risk ecological zones. The project, in its entirety, highlights the fusion of embedded systems, machine learning, and wireless communication into a cohesive, purpose-driven application that addresses one of the most pressing safety challenges faced by tribal populations today. As a whole, the research lays the groundwork for future interdisciplinary advancements in intelligent surveillance systems, offering both social impact and technological contribution in the realms of wildlife conservation, rural safety, and real-time monitoring.

5.2 FUTURE ENHANCEMENTS AND SCALABILITY

While the current implementation of the Tribal Safety and Wildlife Threat Detection System demonstrates a highly effective, real-time threat recognition and alert mechanism suited for remote tribal regions, it also opens avenues for significant future enhancements aimed at increasing its accuracy, autonomy, and large-scale deploy ability. As the system evolves, the incorporation of next-generation deep learning architectures such as YOLOv11, Vision Transformers, or multi-scale attention-based models offers the potential to vastly improve classification accuracy, especially in highly occluded, low-contrast, or multi-animal scenarios commonly encountered in dense forest environments. These

advanced models, when paired with edge accelerators like Google Coral TPU or NVIDIA Jetson Nano, can bring high computational throughput to the edge without increasing power consumption, thus enabling real-time inference on more complex datasets while preserving energy efficiency.

Scalability across extended forest terrains requires not just computational intelligence but also infrastructural adaptability. Enhancing the power subsystem with larger solar panels, high-capacity batteries, and intelligent power management algorithms—including predictive sleep and wake cycles based on environmental triggers—can substantially increase system uptime and resilience in power-constrained settings. To address limitations in communication range and reliability, future deployments can integrate LoRa mesh networking capabilities, where multiple sensor nodes act as relays to ensure robust data transmission over kilometers of terrain without relying on centralized infrastructure. This would allow for real-time coverage of wide forest belts and seamless integration with ranger stations and command centers.

Furthermore, by embedding GPS modules and integrating cloud-enabled dashboards, the system could offer location-aware threat mapping and remote monitoring capabilities, thus equipping forest authorities with actionable intelligence and historical analytics. Cloud synchronization could enable centralized control, multi-site analytics, and integration with other conservation technologies like camera traps and satellite monitoring systems. In addition, the introduction of auxiliary sensing modalities such as vibration sensors, passive infrared (PIR) detectors, or mobile surveillance drones could transform the system into a multi-layered, adaptive threat detection network, enhancing both the temporal and spatial resolution of wildlife activity recognition.

From a socio-political standpoint, future deployments must emphasize extended field trials across diverse ecological zones and over multiple seasonal

cycles to evaluate the system's robustness, environmental adaptability, and maintenance overhead in real-world conditions. Collaboration with governmental agencies, tribal welfare boards, and forest departments will be essential for scaling the system into a national rural safety initiative. With strategic improvements in AI capabilities, energy optimization, communication infrastructure, and cloud integration, the proposed system can evolve into a versatile, sustainable, and policy-aligned solution for addressing the complex challenge of human-wildlife conflict in remote regions across the globe.

REFERENCES

[1] S. P. Sahane, N. S. Salunke, V. S. Shirsath, S. S. Shukla, and A. R. Panhalkar, "Early warning system for detection of harmful animals using IoT," 2019.

[2] R. Upadrashta, T. Choubisa, A. Praneeth, G. Tony, V. S. Aswath, P. V. Kumar, S. Kowshik, H. P. R. Gokul, and T. V. Prabhakar, "Animation and chirplet-based development of a PIR sensor array for intruder classification in an outdoor environment," Dept. Electron. Syst. Eng., Indian Inst. Sci., Bengaluru, Apr. 13, 2016.

[3] G. K. Verma and P. Gupta, "Wild animal detection using deep convolutional neural network," in *Proc. 2nd Int. Conf. Comput. Vis. Image Process.*, 2018.

[4] A. P. N. Achyutha, C. B. C. Chaithra, A. M. Ajay, C. B. Ram, and B. N. Bhavana, "An ephemeral investigation on forest fire, wildlife preservation, and tribal preparedness using deep learning and IoT," Dept. Comput. Sci. Eng., East West Inst. Technol., Bengaluru, India, Dec. 4, 2023.

[5] M. Knyva, D. Gailius, G. Balčiūnas, D. Pratašius, P. Kuzas, and A. Kukanauskaitė, "IoT sensor network for wild-animal detection near roads," Dept. Electron. Eng., Fac. Electr.Electron., Kaunas Univ. Technol., Kaunas, Lithuania, Nov. 2, 2023.

[6] H. Patil and N. Ansari, "Automated wildanimal intrusion detection and repellent system using artificial intelligence of things," in *Proc. 4th Int. Conf. Adv. Sci. Technol. (ICAST)*, Mumbai, India, May 2021, pp. 1–6.

[7] C. Panagiotopoulos, "Machine learning for animal detection and recognition for European wildlife conservation," M.S. thesis, Hellenic Open Univ., Sep. 2024.

[8] C. N. Krishna, A. Rauniyar, N. K. S. Bharadwaj, S. B. Raj, V. Valsan, K. Suresh, V. R. Pandi, and S. Sathyan, "ECO-Guard: An integrated AI sensor system for monitoring wildlife and sustainable forest management," in *Proc. ICT: Smart Syst. Technol. (ICTCS)*, Mar. 2024, pp. 409–419.

[9] N. Abed, R. Murgun, A. Deldari, S. Sankarannair, and M. V. Ramesh, "IoT and AIdriven solutions for human-wildlife conflict: Advancing sustainable agriculture and biodiversity conservation," *J. Sustainable Futures*, vol. 12, no. 2, pp. 8273– 8277, Feb. 2025.

[10] J. S. Dertien et al., "Mitigating human– wildlife conflict and monitoring endangered tigers using a real-time camera-based alert system," *BioScience*, vol. 73, no. 10, pp. 748– 757, Oct. 2023.

[11] H. Hammouch and A. Raghav, "AI and ML applications in wildlife conservation and forest management: A comprehensive review," *J. Artif. Intell. Conserv.*, vol. 15, no. 3, pp. 123–145, 2024.

[12] B. Somwong, K. Kumphet, and W. Massagram, "Acoustic monitoring system with AI threat detection system for forest protection," in *Proc. 20th Int. Joint Conf. Comput. Sci. Softw. Eng. (JCSSE)*, Jun. 2023, pp. 1–6.

[13] E. K. Ronoh, S. Mirau, and M. A. Dida, "Human-wildlife conflict early warning system using the Internet of Things and short message service," *Int. J. Adv. Comput. Sci. Appl.*, vol. 12, no. 2, pp. 8273–8277, Apr. 2022.

[14] T. S. Muthuraju, K. S. Likhith, P. Rao, K. M. Monish Kumar, and B. Sachin, "Wildlife intrusion detection and prevention in farm fields using IoT technology," in *Proc. Intell. Control, Robot., Ind. Autom. (RCAAI)*, Oct. 2024, pp. 43–54.

[15] P. Dokku, S. Mudrakola, K. K. Dadi, and N. Akula, "Identification of wild animals in forest surveillance cameras," in *Proc. Pervasive Knowl. Collective Intell. Web Social Media (PerSOM)*, Aug. 2024, pp. 229–239.

[16] K. V. Reddy et al., "Edge AI in sustainable farming: Deep learning-driven IoT framework to safeguard crops from wildlife threats," in *Proc. IEEE Conf. Sustainable Comput. Commun. (SCC)*, 2023, pp. 1–6.

[17] G. A. Senthil, R. Prabha, N. Aishwarya, R. M. Asha, and S. Prabu, "An IoT integrated smart prediction of wild animal intrusion in residential areas using hybrid deep learning with computer vision," *EAI Endorsed Trans. Internet Things*, vol. 10, p. 1, 2024.

[18] M. Z. Altobel and M. Sah, "Tiger detection using Faster R-CNN for wildlife conservation," in *Proc. 14th Int. Conf. Theory Appl. Fuzzy Syst. Soft Comput. (ICAFS)*, Jan. 2021, pp. 572–579.

[19] M. Murugesan and A. Mahendiran,"Animal conservation applications for wildlife detection and classification using various machine learning algorithms," in *AIP Conf. Proc.*, vol. 3112, p. 020028, 2024.